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Experiment 1: Mechanical Hysteresis

- **Objective**: To verify the relationship between torque and rotation of metal bars.

- **Apparatus:** Metal bars, Spring balance, Mechanical hysteresis setup.

- **Task:**
  (i) Record the hysteresis curves of two different materials and
  (ii) Plot the Torque vs Angle of Rotation graphs for those two materials

- **Principle:** The relationship between torque and angle of rotation is determined when metal bars are twisted. If forces act on a solid body, it is deformed, e.g., with shear stresses, shear deformations will occur. The Hooke's law range is characterized by the linear relationship between stress and torsion. With solid bodies, there is generally a range adjacent to the Hooke's law range, in which there is no longer a linear relationship between stress and deformation, but in which the deformation is still reversible to some extent. The limit of this range is called the yield point. The deformation becomes plastic if the stresses become greater than the yield point. The deformation of the bar is then not completely reversed, even in the stress-free condition. Since the phenomena of plasticity result from displacements of atoms, temperature and time have an influence.
  According to Hooke's law, the relationship between the shear stress (τ_s) and the angular deformation or shear strain (ϒ) is given by

$$\tau_s = \sigma * \Upsilon$$

  where σ is the shear modulus.

If metals are loaded into the plastic range and the material is allowed to relax, it subsequently finds itself again in the Hooke's law range with a new equilibrium position.

Therefore, within Hooke's Law, if you apply a force ***F*** perpendicular to the arm of length ***r***,

The torque is defined as - $\boldsymbol{\tau = r \times F}$

If the force is applied with an angular deformation θ then

$$\tau = rF\sin\theta$$

For a perpendicular force, $\theta = 90°$

Therefore, the final torque becomes -

$$\tau = rF.$$

Figure 1: Photographs of the experimental setup for measuring the hysteresis of metal bars in torsion: (Left) Initial arrangement and (right) Measuring torque versus angular deflection

❖ **<u>Setup and Procedure</u>:**

1. Arrange the experimental setup as shown in Figure 1. Ensure that the spring balance applies force at right angles to the lever arm, so that a pure turning moment is produced.

2. Fix the given metal rod securely to the top clamp and ensure that it is perfectly vertical with respect to the base plate.

3. Using the base plate adjustment, set the pointer/needle to zero on the angular scale before applying any force.

4. Apply a force perpendicular to the horizontal bar using the spring balance. Ensure the force is applied smoothly and without jerks.

5. Measure and note the applied force for every 15° of angular rotation of the rod.

6. Begin observations from 0°, gradually increasing the applied force and recording readings up to 180° of rotation.

7. After reaching 180°, gradually reduce the applied force and continue taking readings until the force returns to zero, noting the corresponding angles.

8. Reverse the direction of the applied force and again record the force values for every 15° increase in angle up to 180°.

9. Repeat the process of increasing and decreasing force in the reverse direction.

10. Plot the applied moment versus angle of rotation for both loading and unloading cycles. The enclosed loop obtained represents the mechanical hysteresis curve.

11. Repeat the experiment for a different metal rod.

12. Except with steel, the elastic limit is very quickly reached, so that the measurements should be carried out either continuously or interrupted by uniform relaxation intervals.

13. For reproducible curves, the torsion bars must not have any kinks or other deformations.

14. In contrast to magnetic hysteresis, in which the crystal structure of the magnetic material is generally unchanged, in the case of mechanical hysteresis a direct relationship is to be found between deformation and moment as a function of time or temperature.

❖ **Observation:**

S.No.	Rotation θ (Deg)	Force F (N)		Torque τ (N-m)	
		Metal rod 1	Metal rod 2	Metal rod 1	Metal rod 2
1	0				
2	15				
3	30				
4	45				
…	…				
…	…				
….	165				
…	180				
…..	165				
…	150				
…	…				
25	0				
26	-15				
27	-30				
…	…				
…	-180				
…	-165				
…	…				
….	0				
…	15		……	……..	
…	30				
….	….				
….	180				

**Take at least 60 readings (for each metal rod)*

❖ **Result:** Do the calculations and plot the following graph.

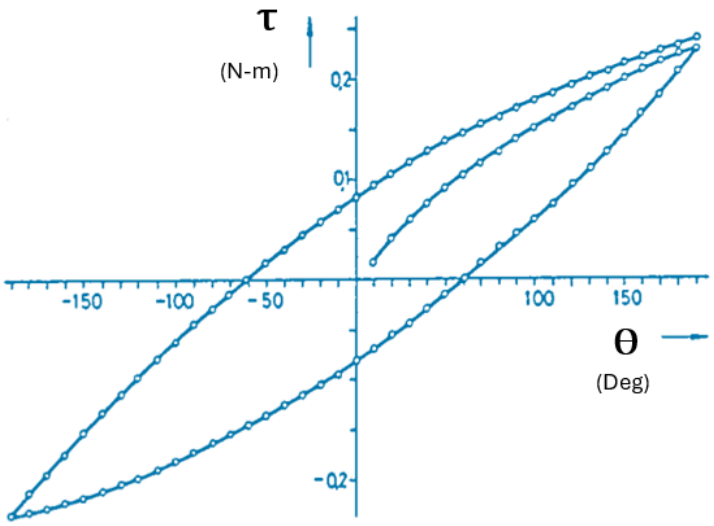


Figure 2: Mechanical hysteresis curve for the torsion of a metal rod.

- **<u>Error Analysis</u>:** Quantify the fractional/percentage error in your results. Do the error calculation using a pen. Discuss among your group members if needed.

  Calculate error using differentiation method –

$$\frac{\Delta\tau}{\tau} = \frac{\Delta F}{F} + \frac{\Delta r}{r}$$

  Where ΔF is the least count of spring balance
  and Δr is the least count of meter scale

- **<u>Source of Error</u>:** Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. etc.

- **<u>Discussion</u>:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 2: Frank Hertz

- **Objective:** To perform the historic Frank-Hertz experiment to demonstrate the existence of discrete energy levels of gas molecule (Neon) and determine the minimum excitation energy of Neon.

- **Apparatus:** Frank Hertz unit, Neon gas tube.

- **Task:** (i) Record the variation in the plate current with change in the potential G2 between the cathode C and the grid G and plot graph.

  (ii) To calculate the value of excitation energy of a Neon atom from a graph.

- **Principle:** The Frank Hertz tube is a tetrode with an indirectly heated Barium Oxide cathode, a mesh type control electrode, a mesh type anode, and a collecting electrode (see figure 1). The electrodes are in a plane parallel configuration. The tube is filled with Neon gas at a pressure chosen to give an optimum characteristic curve

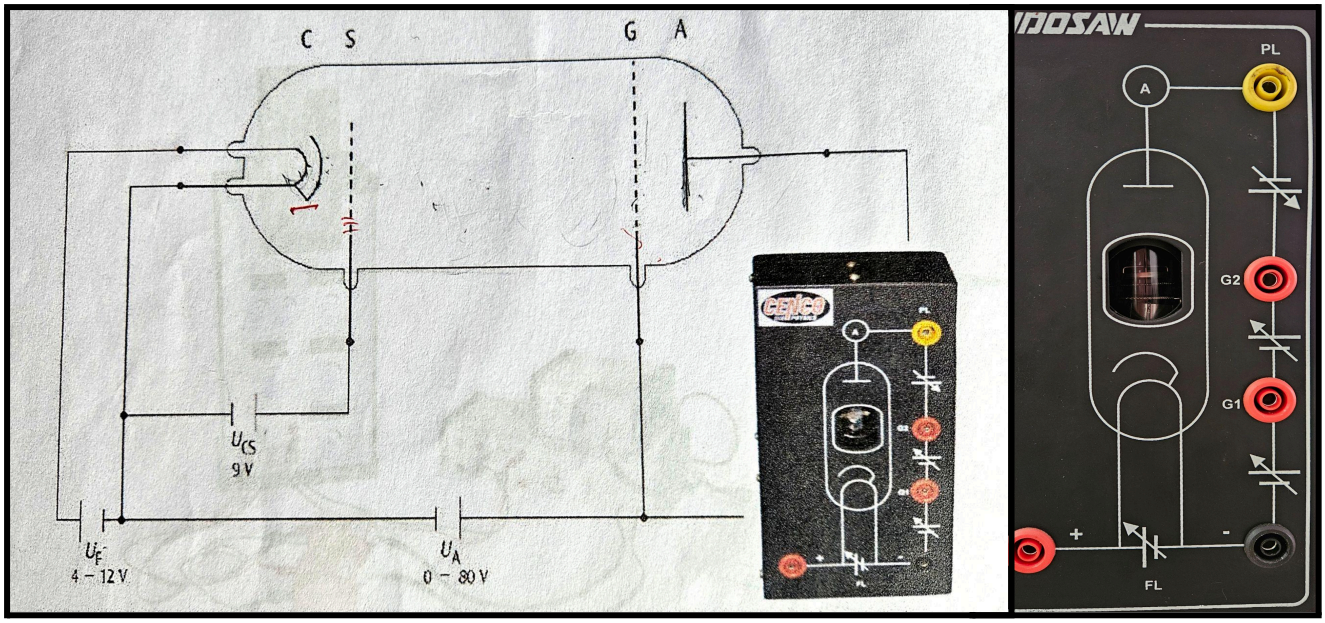


Figure 1: Neon tube schematic diagram

In the Frank Hertz experiment neon atoms are excited by inelastic collision with electrons. The excited atoms emit visible light that can be viewed directly. Thus, it is possible to detect zones where the light and therefore the excitation is more intense. The distribution of such zones between the cathode and the grid depends on the difference in potential between the two.

Electrons are emitted from the cathode and are accelerated by a voltage V towards the grid. Having passed through the grid, they reach the target and thus contribute to a target current if their kinetic energy is sufficient to overcome a decelerating voltage U_{GA} between the grid and the target.

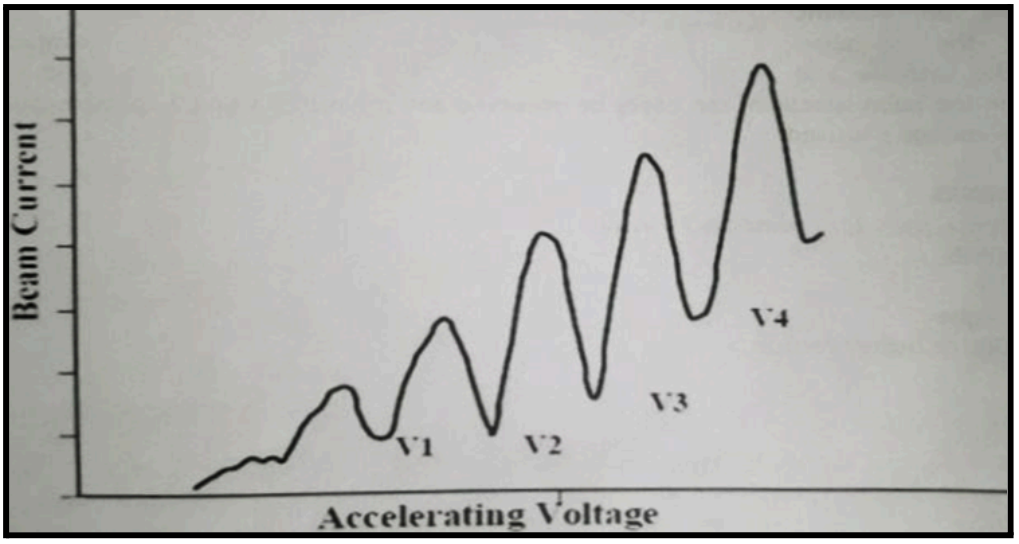


Figure2: The I (U) characteristic has a similar pattern to the original Frank Hertz experiment using mercury gas.

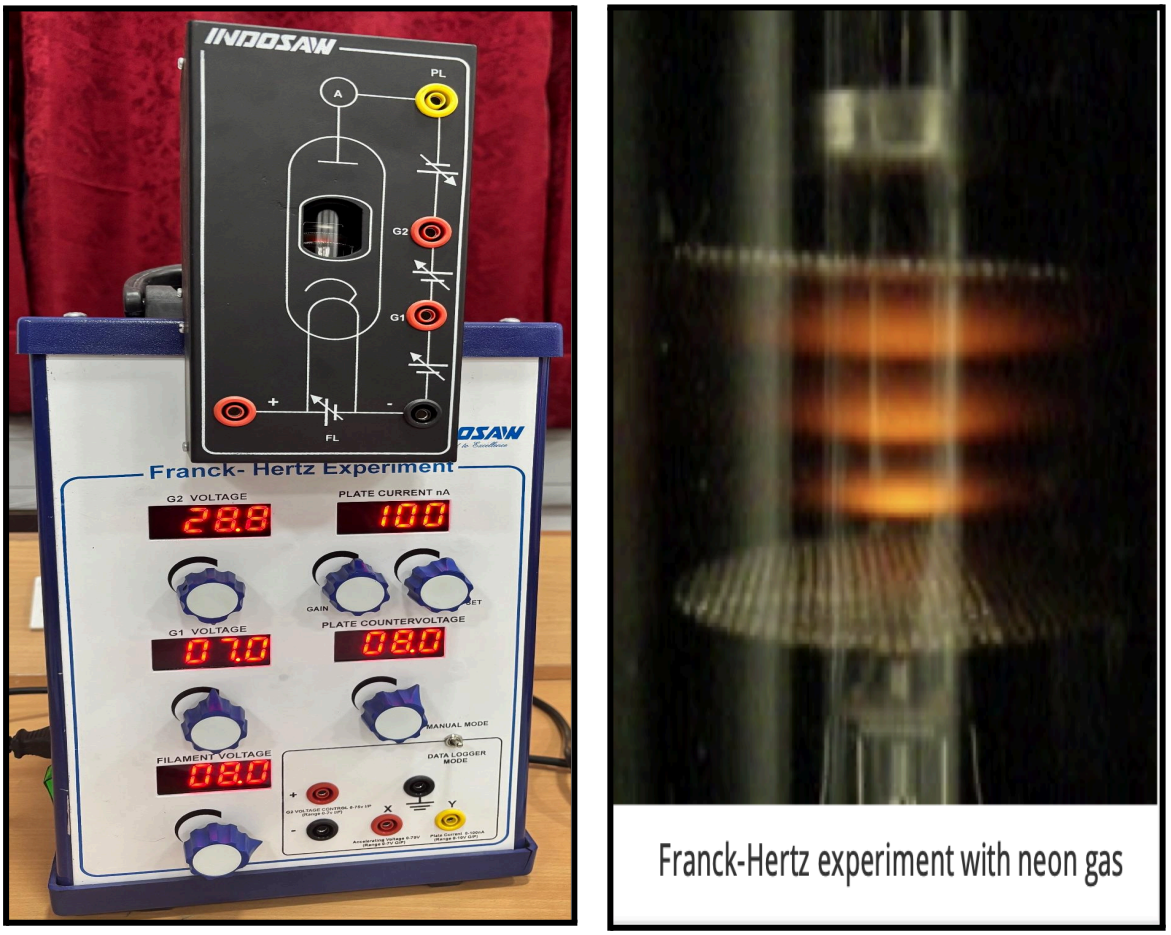


Figure 3: Experimental setup

Technical Details:

- **Filament Voltage:**
- **Control voltage (G1):**
- **Plate counter voltage:**
- **Accelerating Voltage (G2): Maximum 80 V**

Procedure:

(i) Set all the control knob (filament voltage, G1 voltage, plate counter voltage, G2 voltage, gain, offset) at the extreme anticlockwise position of the Frank Hertz base unit.

(ii) Connect the Frank Hertz base unit plug to the mains and switch ON the unit.

(iii) Now set the filament voltage between the given range approx., and wait for 3-5 minutes.

(iv) Set both G1 voltage and plate counter voltage within their respective range (as provided)

(v) Set the GAIN knob at the center position and zero plate current using the OFFSET knob.

(vi) Tune Filament voltage, G1 voltage and plate counter voltage in such a way that you get all three maxima (in plate current value) by slowly increasing G2 Voltage (to ensure you get at least 3 maxima.) You may allow the system to cool down for 10 minutes by switching it off, if it gives increasing values for the third peak. You are now ready to take the final data points.

(vi) For fixed tuned values of Filament, G1 and plate counter voltages, now increase the G2 voltage very slowly (with 1 V or less) and note down the corresponding plate current value. (Take more data points around the peak/maximum in steps of 0.5 V.) Do not exceed the G2 voltage beyond 80 V.

(vii) You must obtain at least 3 peaks.

(viii) If V_n is the potential corresponding to the nth peak and V_1 is the potential corresponding to the 1st peak, then the mean 1st Excitation potential = $(V_n - V_1)/(n-1)$. Where (n-1) is the number of dips between the 1^{st} and the n^{th} peak.

❖ **Observation:**

Sr. No.	G2 Voltage(V)	Plate Current (nA)
1	…..	…..
2	…..	X (1st Maxima)
…….	……	……
…….	…….	……
		X (2nd Maxima)
		……
…….	…….	X (3rd Maxima)
60	……..	……

**Take 60-80 readings (Mark the maxima in the observation table)*

❖ **Result:**

(i) Plot plate current vs G2 voltage curve.

(ii) Calculate the value of the excitation energy of Neon for different peaks and calculate the average Excitation energy. You can calculate the potential difference using peaks (maxima) as well as trough (minima).

The average Excitation energy = $(E \pm \Delta E)$ eV

❖ **Error Analysis:** Quantify the fractional/percentage error in your results.

$$Fractional\ Error = \frac{(Theoretical - experimental)}{Theoretical}$$

$$Percentage\ Error = \frac{(Theoretical - experimental)}{Theoretical}\ x\ 100$$

Theoretical value of Excitation energy of Neon =18.3 eV

Do the error calculation using a pen. Discuss among your group members if needed.

❖ **Source of Error**: Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. Etc.

❖ **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 3: Centripetal Force

- **Objective:** To verify the equation for centripetal force, specifically confirming the functional dependence of centripetal force on mass, radius and frequency.
- **Apparatus:** Centripetal force apparatus, Vernier caliper, stopwatch, slotted weight set, weight hanger.
- **Task:** To calculate and verify the value of centripetal force for different radii.
- **Principle:** The centripetal force apparatus is intended for the study and verification of the law of force for uniform circular motion. It consists essentially of a heavy mass bifilarly supported from the cross arm attached to a vertical shaft (see figure below).

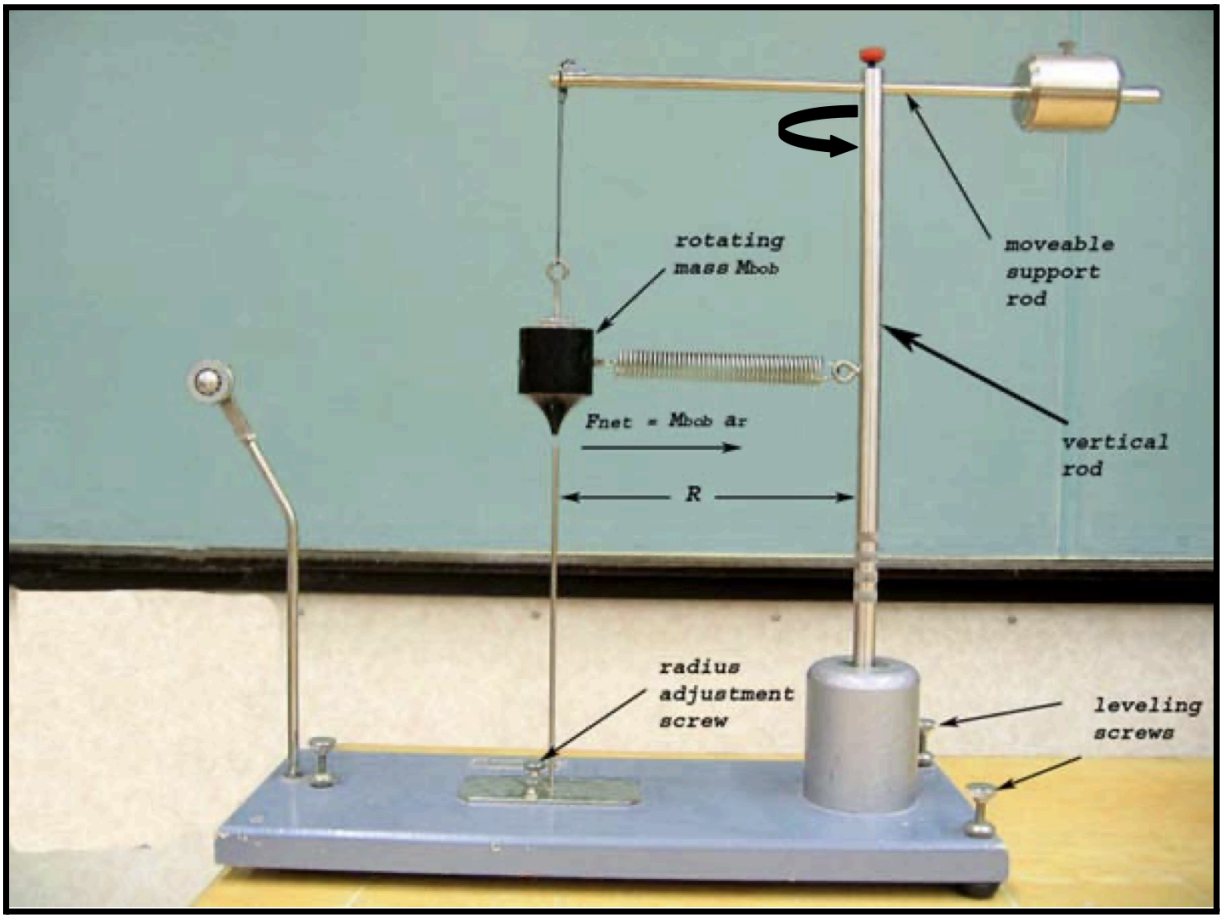


Figure 1: The centripetal force apparatus (Mass Rotating in Uniform Circular Motion)

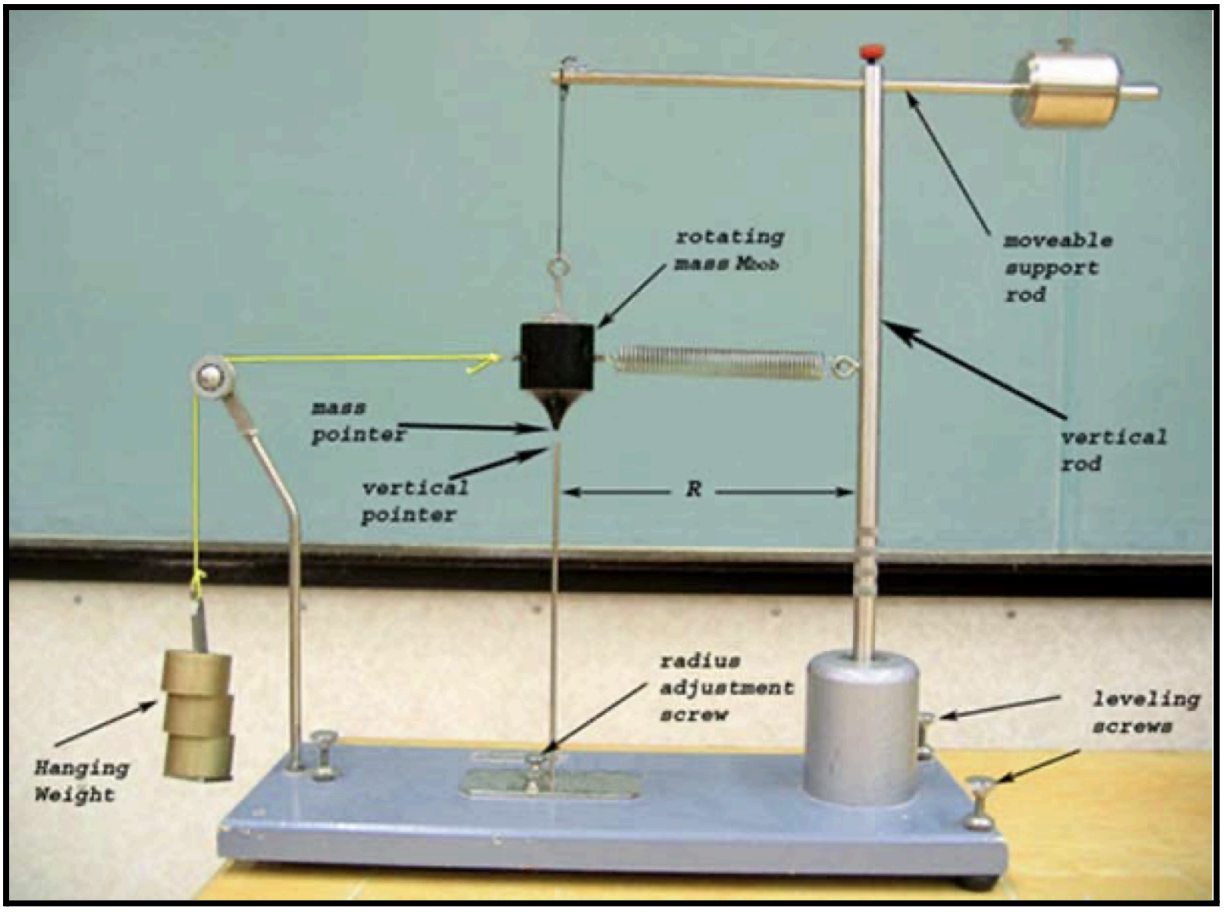


Figure 2: The centripetal force apparatus (Mass in Equilibrium)

When a body is caused to revolve in a circle with uniform velocity, the resultant in-ward force on the body is called "centripetal force". The centripetal force produces an inward radial acceleration (a) given by Newton's Second law

$$F = m*a \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

In which m is the mass of the revolving object.

Since $a = v^2/r$ and $v = 2\pi nr$,

Therefore,

$$F = 4\,\pi^2\, mn^2r \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

For this dynamic configuration, the spring force F provides the Centripetal force. For the static configuration, the spring is stretched by the same amount r by the weight Mg. The spring force is

$$F = Mg$$

The spring force will be the same in both static and dynamic configuration hence

$$Mg = 4\,\pi^2\, mn^2r = mr\omega^2 = mr\,(2\pi)^{\,2}/T^2$$

where r is the radius of circular path, n is the number of revolutions per second. Equation 2 is the working equation for this apparatus. If m is in grams and r is in centimeters, F will be in dyne. If m is in kilograms and r is in meters, F will be in Newton.

❖ **Procedure:**

1. Initial Alignment (**Spring Detached**)

   For each trial, adjust the position of the horizontal cross-arm and the vertical indicator post so that the mass (bob) hangs freely and is directly above the indicator post when the spring is detached.

2. Practice Alignment (**Spring Attached**)

   Attach the spring to the mass. Rotate the apparatus and practice keeping the bottom of the mass (bob) aligned with the indicator post while it is rotating as shown in Figure 1.

3. Timing the Motion

   Measure the time taken for 30 revolutions of the apparatus. Keep the speed as constant as possible by maintaining alignment between the pointer at the bottom of the mass and the indicator post. (Placing a white sheet of paper behind the apparatus can help improve alignment, and sticking **tiny (not thick)** tape at the tip of the vertical post can help count revolutions.)

4. Using the same mass and radius, repeat the timing for three trials and record all data.

5. Calculate Centripetal Force from Motion

   Use the time from the trial to calculate the velocity of the mass. From this velocity, calculate the centripetal force acting on the mass during rotation.

6. Determine Centripetal Force Using Hanging Mass

   With the apparatus at rest, attach a hanging weight to the mass until it is again aligned directly above the indicator post as shown in Figure 2. Because the spring is stretched by the same amount during rotation, and thus the force stretching the spring is equal to the centripetal force.

7. Compare Results

   Calculate the force due to the hanging weight and compare it with the centripetal force obtained in Step 5 by finding the percent difference or fractional error.

8. Repeat for a three different radius (use vernier caliper for better precision to measure radius).

9. Change the radius of rotation and repeat Steps 2 through 7.

10. The following data should be calculated and recorded in your table.

❖ **Observation:**

S.No	Known Mass (kg)	Radius (m)	Time (sec)	Frequency (rev/sec)	Velocity (m/s)	Acceleration (m/s^2)	Calculated force (N)	Hanging Mass (kg)	Measured Force (N)	Fractional error
1	0.445	0.17								
	0.445	0.17								
	0.445	0.17								
2	0.445	…..	…..	……	….	……	…..		…….	
	0.445									
	0.445									
3	0.445	…..	…..	……	….	……	…..		…….	
	0.445									
	0.445									

**Take readings for 3 different radius and take 3 set of reading for each radius.*

❖ **Result:** Value of centripetal force by calculating the average of calculated force.
The calculated average force for a radius is $(F \pm \Delta F)$ N, where ΔF the average fractional error that needs to be calculated as stated in the following section.
Write down the average force for all the three radii.

❖ **Error**: Taking average of fractional errors for each radius, calculate the percentage error for each radius.

❖ **Error Analysis:** Quantify the fractional/percentage error in your results from the average calculated force and measured force. Do the error calculation using a pen. Discuss among your group members if needed.

❖ **Source of Error:** Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. Etc.

❖ **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 4: Magnetic field along the axis of a single coil

- **Objective:** To study the magnetic field along the axis of a circular coil at different positions.

- **Apparatus:** Power Supply, Gauss meter, Axial Probe, Jack, Coils, Meter ruler, Connecting leads.

- **Task:**

  (i) Record the variation of B(x) as a function of x for coil having number of turns (n) = 1150 & 165 at current (I) = 0.5A & 0.7A and plot curve.

  (ii) Record the variation of log (B(x)) as a function of log (R^2+x^2) for n = 1150 coil at current (I) = 0.5A & 0.7A and plot curve. From graph, find the value of permeability of air (μ_0).

- **Principle:** The Biot–Savart law is an equation that describes the magnetic field generated by an electric current. This law is used for computing the resultant magnetic field B at position r generated by a steady current I.

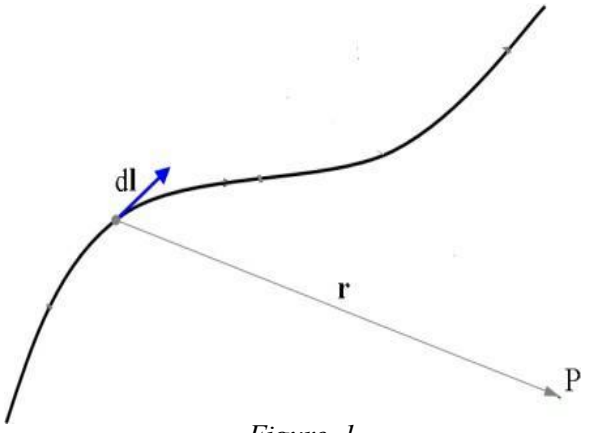


Figure. 1

According to the law, the magnetic field (dB) for an infinitesimal part of a conductor (dl), carrying current I (fig. 1), at a distance r is determined by the following formula:

$$dB = \frac{\mu_0}{4\pi}\frac{Idl\,x\,r}{|r^3|}$$

where μ_0 is the permeability of the free space.

As an application, the magnetic field at a distance x along the axis of a circular coil having n turns, carrying a current I, is (calculated by that formula)

$$B(x) = \frac{\mu_0 nIR^2}{2}\frac{1}{(R^2+x^2)^{3/2}}$$

where R is the radius of the coil,

❖ **Experimental Setup:** Set up the experiment as shown in figure 2

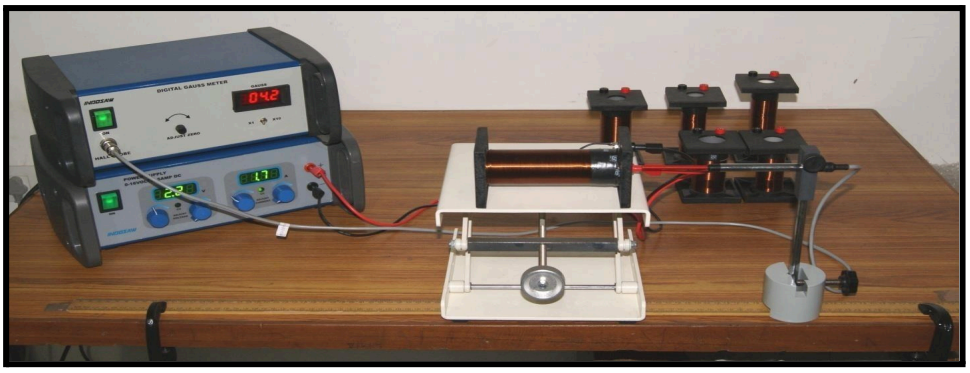


Figure 2: Experimental Setup

❖ **Procedure:**

(i) Connect the DC Power supply and circular coil in series as shown in Figure 2

(ii) Adjust the probe in such a way that it is perfectly aligned along the central axis of the coil throughout the experiment.

(ii) Switch on the DC Supply. Rotate the current and voltage knob to the desired value.

(iii) Note that the current should be constant throughout the experiment.

(iv) You are provided with an Axial Hall Probe connected to a Gauss meter. The probe's end portion detects the magnetic field. Move the probe along the center of the coil and check the position at which the magnetic field is maximum.

(v) Note down the readings of the magnetic field at different positions with a step of 1 cm **(check alignment of the Axial Hall probe for every reading/1 cm)**. The magnetic field in the Gauss meter fluctuates due to some electrical noise. At each step, wait for a few seconds and note down the average value.

(vi) Draw the graph between distance and magnetic field due to coil as shown in figure 3

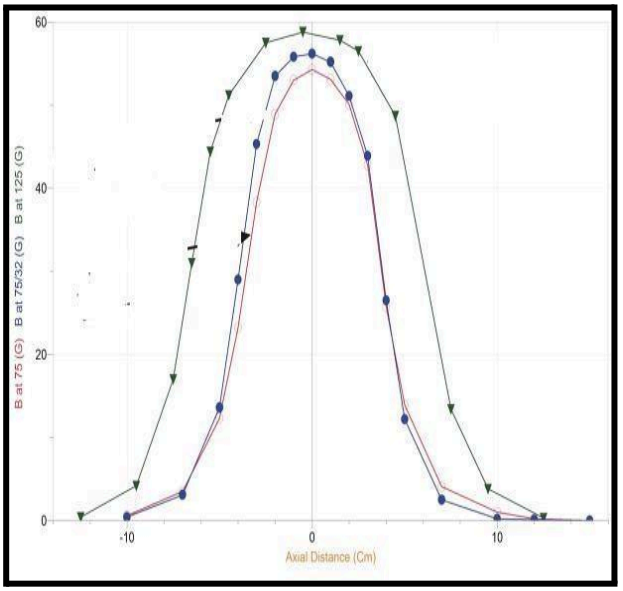


Figure 3: The change of the magnetic field along the axis of the coil.

❖ **Observations:**

Observation 1:

(i) Radius of the coil = ….
(ii) x_0 = ….

No. of turns in the coil	Current flowing through the coil (A)	Probe position (x) (cm)	Distance X= ($x-x_0$) (cm)	Magnetic field (B(x)) (Gauss)	log(B(x))	log (R^2+x^2)
1150	0.5	x1				
		x2				
		.				
		.				
		x15				
	0.7	x1				
		x2				
		.				
		.				
		x15				

* *Take a 15-20 set of readings for each coil*

Observation 2:

(i) Radius of the coil = ….
(ii) x_0 = ….

No. of turns in the coil	Current flowing through the coil (A)	Probe position (x) (cm)	Distance X= ($x-x_0$) (cm)	Magnetic field (B(x)) (Gauss)
165	0.5	x1		
		x2		
		.		
		.		
		x15		

❖ **Result:**

(i) Plot of B(x) vs X on a single graph sheet for the current 0.5 A for two different coil turns.

(ii) Plot of B(x) vs X on a single graph sheet for the current 0.5 A and 0.7 A for coil turns to be 1150.

(iii) Plot log(B(x)) vs log (R^2+x^2) for the Table 1 data for two different current 0.5 A and 0.7 A [Note: make sure the parameters are following the same unit system]

(iv) Find out the value of the permeability of air (μ_0) using the intercept of the log(B(x)) vs log (R^2+x^2) plots. [Note: make sure the parameters are following the same unit system]

[#NOTE: Do not forget to mention the scales for both X and Y axes with units in the graph paper for each plot]

The permeability of air (μ_0) = $\mu_0 \pm \Delta\mu_0$ N/A^2.

❖ **Error Analysis:** Quantify the fractional/percentage error in your results. Do the error calculation using a pen. Discuss among your group members if needed.

Theoretical value of μ_0 = $4\pi \times 10^{-7}\ \mathrm{N/A^2}$

❖ **Sources of Error**: Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. etc.

❖ **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 5: Stefan's Constant

- **Objective:** To determine the value of Stefan's constant using blackbody radiation.

- **Apparatus:** Black body radiation enclosure with blackened silver disk D connected to Ag-Cn, Thermometers, Suspended coil galvanometer, Thermocouple (Cu-Cn), Steam generator, Spirit lamp, Oil bath, water bath, connecting wires.

- **Task:**

  (i) To study the calibration of the given Cu-Cn-Cu thermocouple by measuring the rate of change of temperature.
  (ii) To calculate the value of Stefan's constant.

- **Principle:** The apparatus consists of a hollow metallic sphere blackened from inside and placed inside a rectangular box which serves as steam chamber. The whole arrangement rest on a wooden box which serves as a steam chamber. The whole arrangement rest on a wooden plate which has small hole 2 cm in diameter exactly at the center of the hemisphere. A small disk D whose upper surface is coated with lamp black can be put in position into the hole with the help of the ebonite rod R fixed to it. The silver disk D can be covered and uncovered from the radiation of the enclosure by moving ebonite rod up and down using the screw. This can be arranged from outside with the help of the handle.

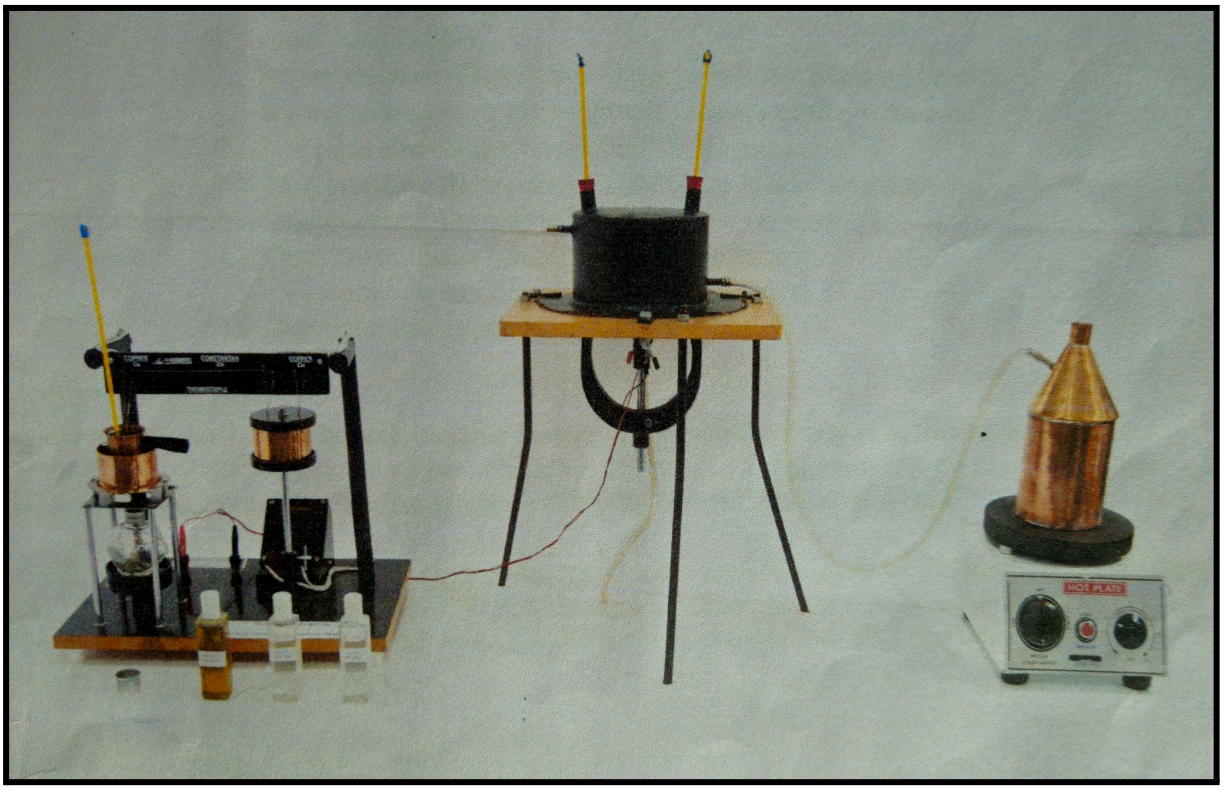


Figure 1: Picture of the apparatus.

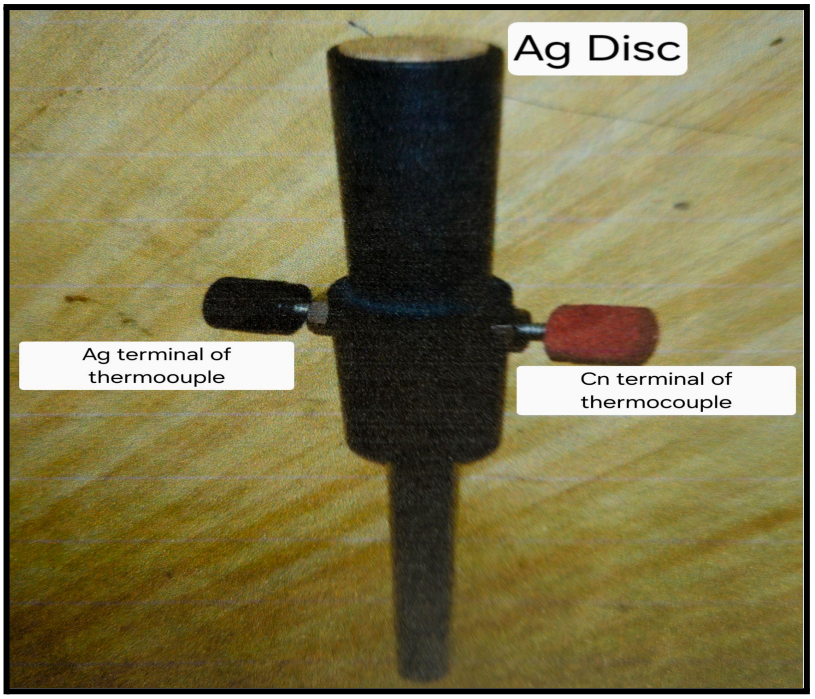


Figure 2: Blackened Silver disc (D) or Ag Disc

Let R_1 and R = Radiation absorbed and emitted by the silver disk per unit area per second respectively
Let the temperature of surrounding = T_1
Temperature of disk = T_2
Mass of disk = m
Specific heat of disk = s
Time rate of change of temp. = dT/dt
Then,

$$ms\frac{dT}{dt} = \frac{\sigma A}{J}(T_1^4 - T_2^4)$$

Or

$$\sigma = \frac{Jms}{A(T_1^4 - T_2^4)}\frac{dT}{dt}$$

Stefan's Law state that rate of loss of heat from a black body per unit time is directly proportional to 4th power of temp of black body
i.e. $E \propto T_1^4$
or $E = \sigma T_1^4$
where σ is called Stefan's constant.
If the body is surrounded by absolute temp. T_2 then,

$$E = \sigma [T_1^4 - T_2^4]$$

If the body is a non-black body, then

$$E = \epsilon \sigma [T_1^4 - T_2^4]$$

where Є = emissivity of non-black body and is always less than one.

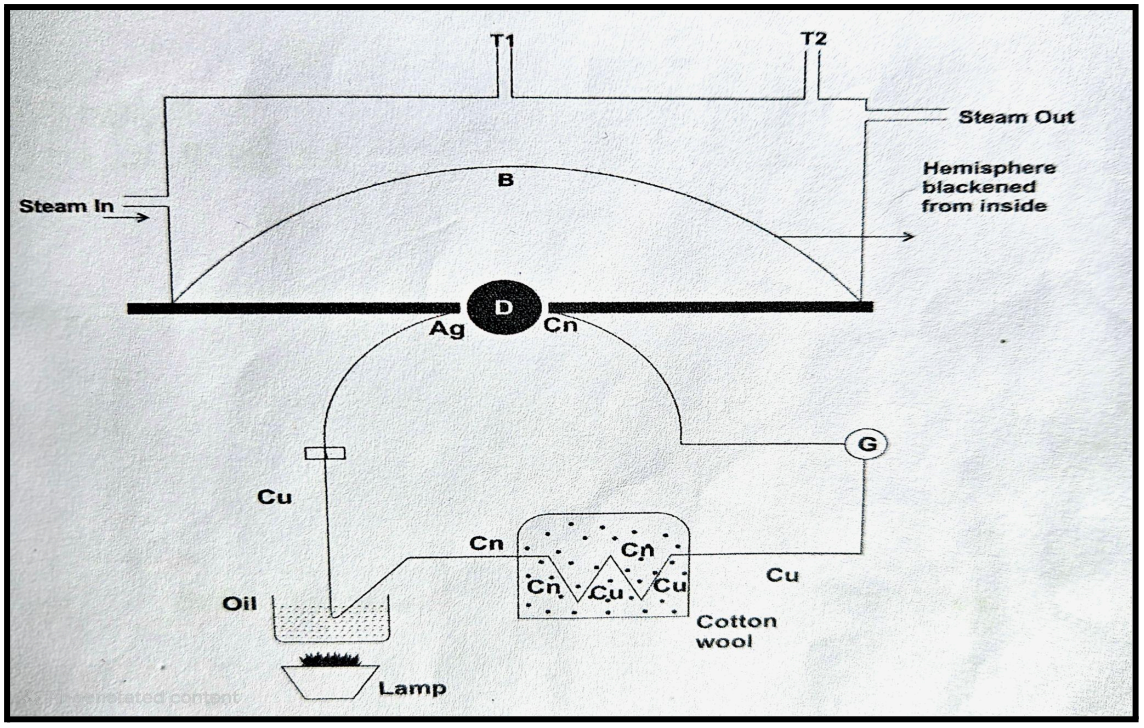


Figure 3: Experimental arrangement for the determination of Stefan's Constant

❖ **Procedure:**

Determination of dT/dθ:

(i) Set the silver disk (D) into a hole as shown in figure 2 by moving the rod of disk (D) upwards (use the middle screw to loosen/tighten the rod of disc). This will treat a silver disk (D) placed at the cold junction.
(ii) Heat the calorimeter C_1 by placing it into the mustard oil bath using a spirit lamp. This will produce a temperature difference between calorimeter C_1 and disk (D) using the two thermojunctions of thermocouple.
(iii) Observe the thermoelectric current by the Galvanometer fitted on board.
(iv) Note down the temperature of the calorimeter C1 (hot junction) containing mustard oil and corresponding deflection in the Galvanometer.
(v) Take readings from a room temperature up to 150 ^0C *, for every* 5^0C *temperature rise (it takes up to 1 hour to reach maximum temperature so start this step immediately after reaching the lab)*
(iv) Plot the graph between the temp difference of the two junctions and the corresponding deflection.
(v) From the straight-line graph as shown in graph 1, find slope

$$PQ/QR = \frac{dT}{d\theta} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (1)$$

Note: During this part, disc D acts as a cold junction of the thermocouple.

Determination of dθ/dt:

(vi) Pull out the disc (D) completely from the hole, then remove the spirit lamp and replace the oil bath with the water bath. (now the water bath acts as a cold junction)

(vii) Connect the tube to the inlet of the black body chamber and turn on the steam generator heater (you may keep heater at the maximum setting).

(vii) This will pass the steam inside the chamber, till steady temperature is achieved as indicated by the two thermometers. The temperature in both should be near to 100 ^0C *(it takes up to 50 minutes to reach* steady *temperature in both the thermometer)*

(viii) Push the disc (D) again in the hole inside the black body chamber to absorb radiation from the radiator B (be ready to take readings immediately after inserting the disc).

(ix) The disc now acts as a hot junction. Note down the time corresponding to each deflection of the galvanometer i.e. starting from 1,2,3…

(ix) Draw the graph between the increasing deflection and time (as shown in graph 2).

(x) Draw the tangent at point N on the curve (i.e. at the maximum curvature).

(xi) The deflection θ at point N in graph 2 corresponds to the temperature (T) value in $\frac{dT}{d\theta}$ in graph 1. Now find the slope/value of $\frac{d\theta}{dt}$ from graph 2

$$\text{GH/HL} = \frac{d\theta}{dt} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

From equations (1) & (2) we get -

$$\frac{dT}{dt} = \frac{dT}{d\theta}\frac{d\theta}{dt} \ldots\ldots\ldots\ldots\ldots (3)$$

Determination of T_2:

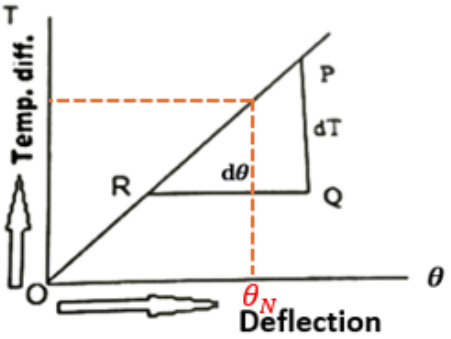
(xi) Corresponding to point N from graph 2 note the value of galvanometer deflection and then note the difference in temperature (t_N) between two junctions from graph 1
(xii) Add the initial temperature of disc to the temperature difference of step (xi) to find T_2 (^0C)
(xiii) Convert T_2 into kelvin by adding 273

- **Observation:**

Observation 1: Galvanometer deflection with the temperature difference
Room Temperature = T_0 = (Kelvin)

Sr. No.	Temp. of the hot junction (^0C)	Temp. of the hot junction T (Kelvin)	Temp difference $T - T_0$ (Kelvin)	Galvanometer deflection Θ
1	30			
2	35			
3	40			
4	….			

** Take readings from room temperature up to 150 ^0C (25-30 set of readings)*

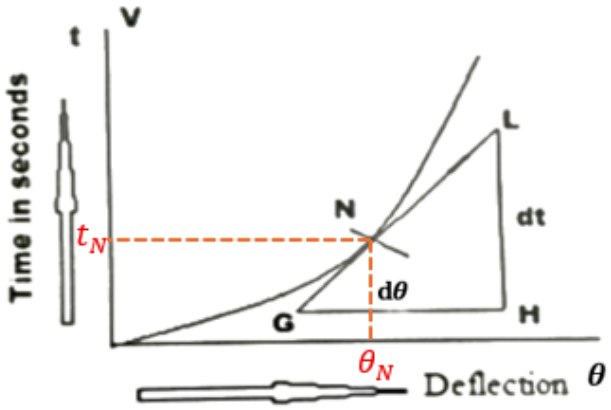


Graph 1: To find $\frac{dT}{d\theta}$

Observation 2: Galvanometer deflection with time

Sr. No.	Galvanometer Deflection Θ	Time t (sec)
1	1	……..
2	2	…….
3	3	……
……..	…….	……

**Take 20-30 set of readings*



Graph 2: To find $\frac{d\theta}{dt}$

Formula used: - $\sigma = \frac{Jms}{A(T_1^4 - T_2^4)} \frac{dT}{dt} = \frac{Jms}{A(T_1^4 - T_2^4)} \frac{dT}{d\theta} \frac{d\theta}{dt}$ **erg s^{-1} cm^{-2} K^{-4}**

where
m = mass of silver disk = 2 gm
s = specific heat of disk = 0.056 cal/g/k
J = mechanical equivalent of heat = 4.2 x 10^7 erg/cal
r = radius of disk = diameter/2 = 2/2 = 1 cm
A = area = πr^2
T_1 = steady temperature of steam
T_2 = (Initial temp of disk i.e., room temperature + temp. difference for deflection (t_N))
$\frac{dT}{dt} = \frac{dT}{d\theta} \frac{d\theta}{dt}$ = from equation (3)

Theoretical value of σ = 5.67 x 10^{-5} erg s^{-1} cm^{-2} K^{-4}

❖ **Result:**

(i) Plot Temperature difference versus Delflection and find out $dT/d\Theta$ from the slope (Graph 1)
(ii) Plot Time versus Delflection and find out $d\Theta/dt$ from the slope drawn at the point of maximum curvature (Graph 2)
(ii) Calculate the value of Stefan's constant σ.

The Stefan's constant is ($\sigma \pm \Delta\sigma$) erg s^{-1} cm^{-2} K^{-4}.

[#NOTE: Do not forget to mention the scales for both X and Y axes in the graph paper for each plot]

❖ **Error Analysis:** Quantify the fractional and percentage error in your results. Do the error calculation using a pen. Discuss this with your group members if needed.

Theoretical Value of σ = 5.67 x 10^{-5} erg/s/cm^2/K^4

❖ **Source of Error:** Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. etc.

❖ **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 6: Magnetic Induction

- **Objective:** To verify Faraday and Lenz's law of induction by measuring the induced voltage as function of time.

- **Apparatus:** Voltage Sensor, Coil, Tube, Digital timer, Photogates, Support base, Support rod, Universal clamp, Einstein labmate.

- **Task:**
  (i) Record the variation of induced emf (U) as a function of falling time and plot curve.
  (ii) Record the variation of peak to peak voltage (U_{pp}) and Total flux (ϕ) with velocity of the falling magnet by changing the height.
  (iii) Plot U_{pp} vs velocity & ϕ vs velocity.

- **Principle:** A permanent magnet falls with different velocities through a coil. The change in the magnetic flux (ϕ) generates an induced voltage impulse. The induced voltage impulse (U) is recorded with a computer interface system. Depending on the polarity of the permanent magnet the induced voltage impulse is negative or positive.

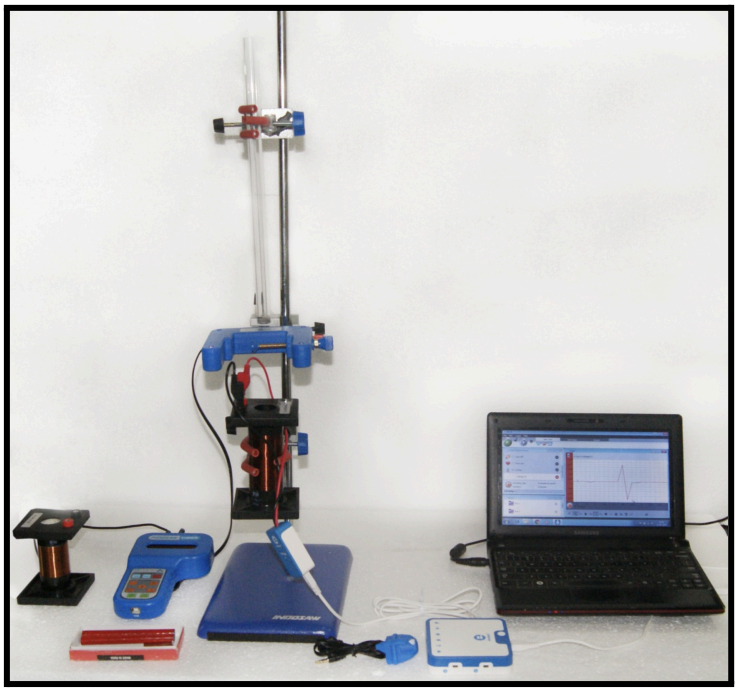


Figure 1: Photograph of the experimental setup

Electromagnetic induction: The phenomenon of production of a potential difference (or current flow) across (or in) a conductor when exposed to a varying magnetic field. Faraday's law of electromagnetic induction explains the working principle of most of the electrical motors, generators, and electrical transformers. This law predicts how a magnetic field will interact with an electric circuit to produce an emf.

A conducting loop which has voltage or current probe attached to it will register a voltage or current if the magnetic flux through the loop changes in time. The change may arise from the motion of a magnet. The emf induced in a loop is proportional to the rate of change of magnetic flux through it

i.e. $$U = -N\frac{d\phi}{dt}$$

where N = Number of windings in the coil, ϕ = Magnetic flux, t = Time

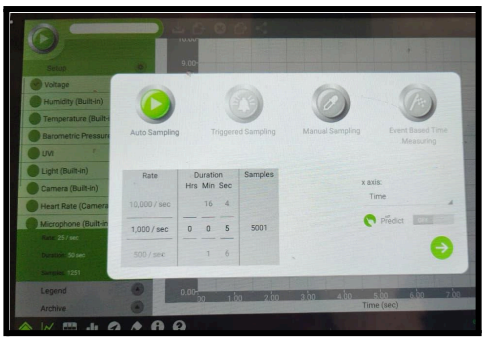
Lenz's law: This law states that the induced emf (and current) will be in a direction such that it opposes the origin of the flux change.

❖ **Procedure:** The experimental setup is arranged as shown in Figure 1.

(i) First check the alignment of the tube with the coil and rest of the connections very carefully. Collect the necessary items to perform the experiment.

(ii) Switch on the digital timer and set MODE>SPEED>SELECT>MASK>PRESS START/STOP KEY. Set the length of the magnet by the up arrow key. Press START/STOP KEY to record the speed of the falling object directly.

(iii) (a) To record the data if using the 'Tab' then open the MiLAB/Logger Pro software



➔ Go to the 'Set-up' icon and set the
- ◆ Duration' to 5-10 seconds,
- ◆ 'Rate' to 1000 /sec and
- ◆ 'Samples' will be Automatic

➔ Use Run/Pause button in the leftmost corner to control the measurement time while dropping the magnet through the tube. You get the voltage vs time series data as shown in Figure 2. Note down the data.

(b) To record the data if using the 'Desktop', then open the MiLAB/Logger Pro software

➔ Go to the 'Set-up' icon (for some system; Experiment > Data Collection) and set the
- ◆ 'Sampling' as AUTO,
- ◆ 'Rate' to 1000 samples per sec,
- ◆ 'Samples' to 5000
- ◆ 'Duration' to 5-10 seconds

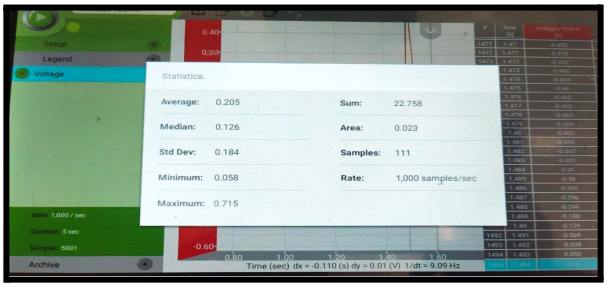
➔ Use Run/Pause button in the leftmost corner to control the measurement time while dropping the magnet through the tube. You get the voltage vs time series data as shown in Figure 2. Note down the data.

(iii) Introduce the tip of the magnet in the glass tube, let it fall; catch it with one hand under the coil. Note down both the velocity data and Voltage vs time data.

(iv) Make sure you catch the magnet every time without letting it fall on the desk/ground. Read the velocity of the falling magnet in the timer. Verify that 'velocity' data with the calculated velocity using $v = (2gh)^{1/2}$ **[use a ruler and a separate digital timer to measure the 'height'**

and the 'time to fall', respectively].

(v) Mark the points a and b as shown in Figure 2 (Right), If using the 'Tab', then go to 'Statistics' and measure the total amplitude U_{pp} (peak to peak) from the maximum and minimum. Also note down the area. Another way to get maximum/minima is by just clicking the points on the obtained graph.



If using the 'Desktop', go to Analysis > Statistics. Note down the Maximum, minimum and the area information.

(vii) Mark the positive (F1) and negative (F2) part of the curve separately and calculate the area with the show integral icon. Finally, add the values of the two partial areas and calculate the flux as shown in Figure 2 (Left).
Alternatively, you can mark two points a and b (as shown in Figure 2 (Right)) and get the total area under the curve at once.

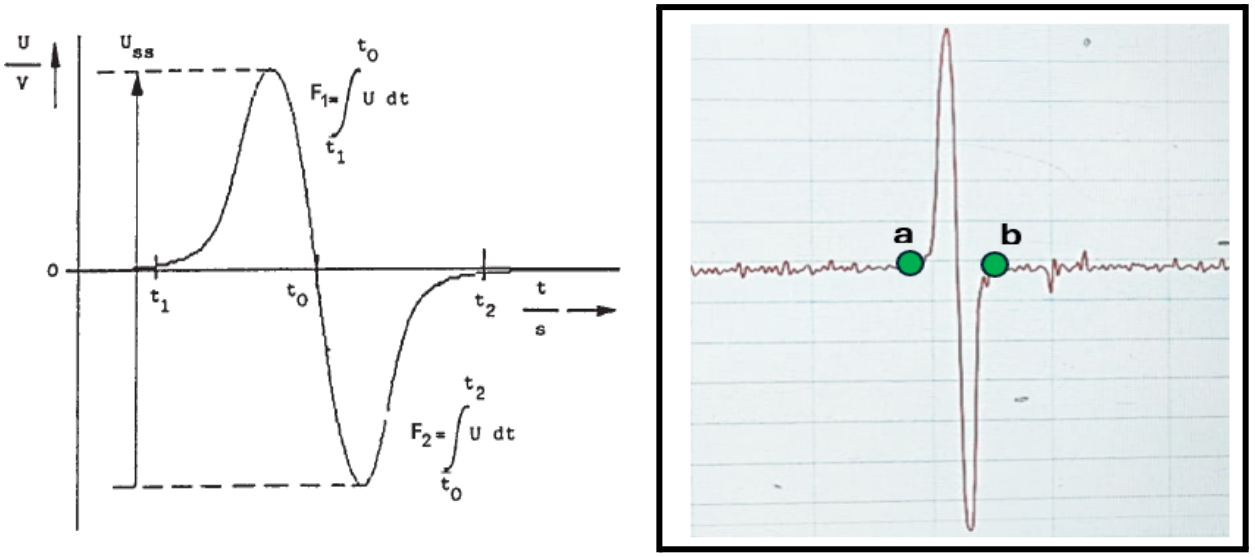


Figure 2: Induced emf (U) vs Time (t)

❖ **<u>Observation</u>:**

Height =

Table 1

S. No.	Induced Voltage U (Volts)	Time t (sec)
1		
2		
3		
…….	……	…….

**Take 20-25 set of readings*

S. No.	Height (m)	Velocity (m/s)	Peak-to-peak value of the induced voltage U_{pp} (Volts)	Magnetic flux Φ (Wb)
1	h1			
2	h2			
3	h3			
…….	……	…….	……	…….

Table 2 **Take readings for 10 different heights. Take a minimum of 4 cm of variation between the different heights.*

- **Calculations:** According to Faraday's law of induction, the induced voltage is given as -

$$U = - N \frac{d\phi}{dt}$$

where N = Number of windings in the coil, ϕ = Magnetic flux, t = Time

On integration,

$$\phi = -\frac{1}{N}\int U dt; \quad \text{where, U = induced voltage and } \int U\, dt = F_1 + F_2$$

- **Result:** (i) Plot of induced emf/voltage (U) vs time.
  (ii) Plot of peak to peak voltage (U_{pp}) vs velocity.
  (iii) Plot of total flux (ϕ) vs velocity.

- **Error Analysis:** Quantify the percentage error in your results (by propagation of error using the differentiation). Do the error calculation using a pen. Discuss among your group members if needed.

$$\frac{d\phi}{\phi} = \frac{dF1}{F1} + \frac{dF2}{F2}$$

- **Source of Error**: Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. etc.

- **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 7: Magnetic field in a Helmholtz coil

- **Objective:** To study the magnetic field along the axis of a Helmholtz Coil for different separation of coils and superposition of field.

- **Apparatus:** Power Supply, Gauss meter, Coil having N = 390 & Dia = 150 mm, Axial Probe, Support base, Support rod, Connecting cables, Probe holder.

- **Task:**
  (i) To measure the magnetic flux density along the x-axis of the circular coils at different positions when the distance between them is a = R, a < R & a > R, where R = radius of coil
  (ii) plot B(x) as a function of x for different a
  (iii) Calculate the magnetic field strength B at the mid-point of the coils when a = R.

- **Principle:** The magnetic field of the coil arrangement shown is rotationally symmetrical about the axis of the coils, which is chosen as the x-axis of a system of cylindrical coordinates (x, r, ϕ). The origin is at the centre of the system. The magnetic flux density does not depend on the angle ϕ, so only the components B (x, r) are measured.
  We computed the field on the axis of a circular current loop. When a current I exist in a loop of radius R, the magnetic field strength B along an axis through the centre of the loop and a distance x from the centre of the loop is given by

$$B = \frac{\mu_0 I}{2} \frac{R^2}{(R^2+x^2)^{3/2}} \quad \text{(1)}$$

You will find in the derivation of Equation 1 with the assumption that the origin is located at the loop centre. If the origin does not coincide with the centre of the loop, but is instead a distance x_0 from the origin, we must replace x in Equation 1 by $(x - x_0)$. Moreover, if the loop actually consists of N turns, each carrying a current I_0, we can replace the current I by NI_0.

With these substitutions, equation 1 becomes

$$B = \frac{\mu_0 N I_0}{2} \frac{R^2}{[R^2+(x-x_0)^2]^{3/2}} = \frac{\mu_0 N I_0}{2} \frac{R^2}{[R^2+(d)^2]^{3/2}} \quad \text{(2)}$$

This equation can be written more compactly by defining B_0 as the field at the centre of the loop (at which $x = x_0$, i.e., d=0.0)

$$B_0 = \frac{\mu_0 N I_0}{2R} \quad \text{(3)}$$

So that,

$$B = B_0 \frac{R^3}{[R^2+(x-x_0)^2]^{3/2}} = B_0 \frac{R^3}{[R^2+(d)^2]^{3/2}} \quad \text{(4)}$$

Helmholtz coils are constructed from two circular coils of wire, each perpendicular to the same axis, and each carrying the same current in the same direction. As shown in Figure 1, the coils are separated by a distance R, which is also the radius of each coil.

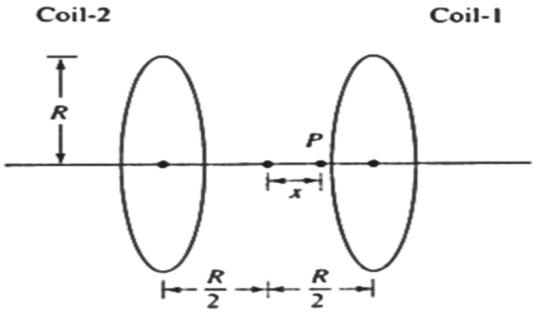


Figure 1

We can use Equation 4 to find an expression for the B field at any point P on the axis of the coils. If the magnetic field strength due to coil 1 is B1 and that due to coil 2 is B2, then by superposition

$$B = B1 + B2 \ldots\ldots\ldots\ldots\ldots\ldots (5)$$

In this configuration, it is convenient to specify x_0 not at the centre of a single coil, but rather at the midpoint between the two coils with respect to the new origin at the mid-point of the two identical coils.

❖ **<u>Procedure</u>:**

(i) Connect the DC Power supply and circular coil in series as shown in Figure 2

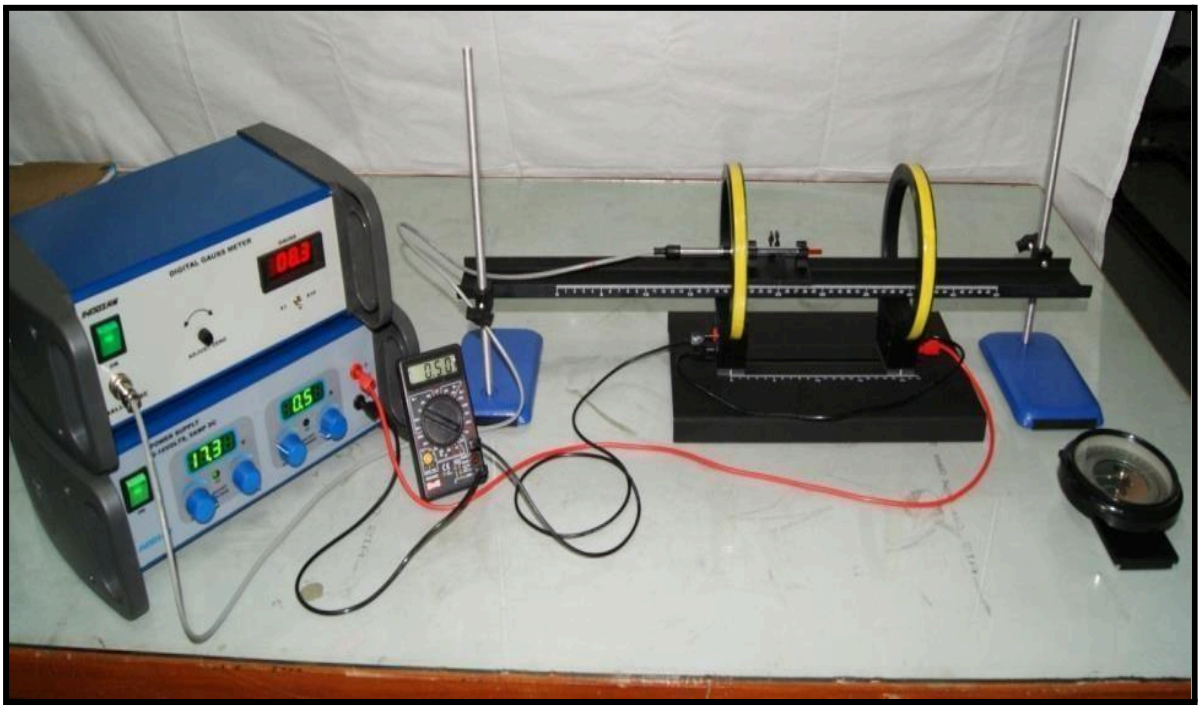


Figure 2: Photograph of the experimental setup

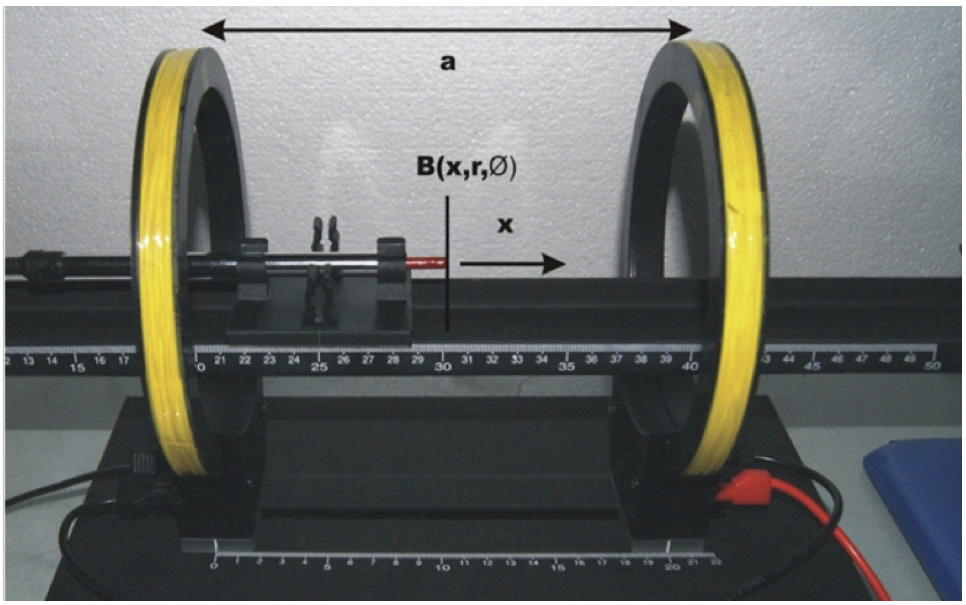


Figure 3: Magnetic Field B(x) along the x-axis and center at $x = x_0$

(ii) Switch on the DC Supply. Rotate the current and voltage knob to the desired value.

(iii) Note that the current (0.5 A) should be constant throughout the experiment.

(iv) Set the distance between two coils a = 2R (or R/2 or R) (as shown Figure 3)

(v) You are provided with an Axial Hall Probe connected to a Gauss meter. The end portion of the probe detects the magnetic field. Ensure the probe is aligned exactly along the axis of the coils.

(vi) Move it to the center of the two coils and check the position at which the magnetic field is maximum (this position is x_0).

(vii) Take readings of magnetic field at different positions in steps of 1 cm (on both sides of the x_0 or maxima), without disturbing the setup and keeping the probe along the axis of the coil. Make sure the probe is aligned along the axis for each reading. **(check alignment of the Axial Hall probe for every reading/1 cm).**

(viii) Repeat the experiment for different coil separation. Take minimum 40 set of readings and make sure axial distance range is from **-20 cm to +20 cm** as shown in the figure below.

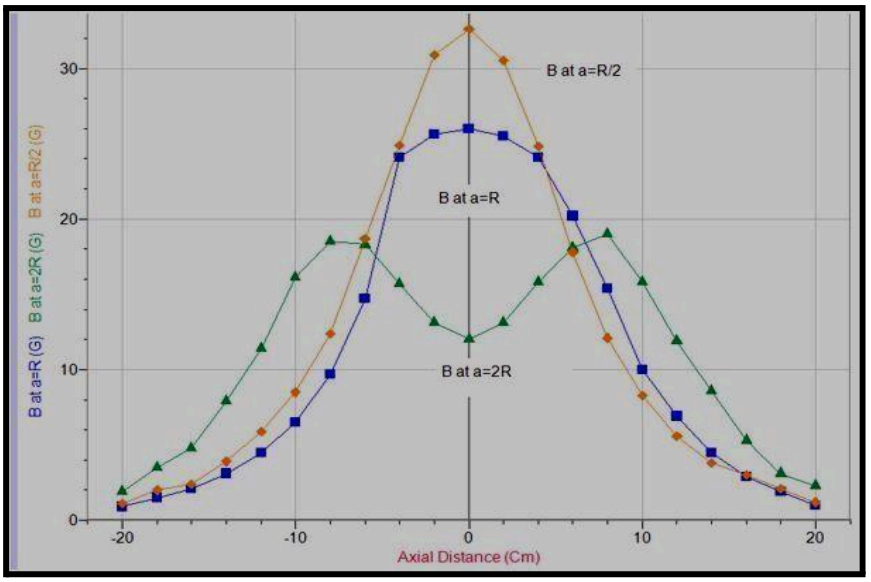


Figure 4: Plot of Magnetic Field (B) vs distance (d)

(ix) The magnetic field in Gauss-meter may fluctuate due to the electronic noise. At each step, wait for a few seconds and note down the average value.

(x) Draw the graph of the magnetic field versus axial distance for three different coil separations 'a' on a single graph sheet as shown in Figure 4

- **Observation:**

  $x_0 =$ .....

Serial No.	Probe position X (cm)	Axial Distance $d = x - x_0$ (cm)	Magnetic field B for a = R/2 (Gauss)	Magnetic field B for a = R (Gauss)	Magnetic field B for a = 2R (Gauss)
1					
2					
3					
..					
..	…….				
40	…….	…..	……	……	…….

Take minimum 40 set of readings* (make sure axial distance (d) range is from **-20 to +20 cm*)*

- **Result:** (i) Plot of B(x) vs d (make sure x-axis range is from -20 cm to +20 cm)
  (ii) Calculated Value of B at mid-point when a = R

  Error Analysis: Calculate the error in the magnetic field (B) and quantify the fractional/percentage error in your results. Do the error calculation using a pen. Discuss among your group members if needed.

  Therefore, the value of B at the midpoint is ($B \pm \Delta B$) Gauss

- **Source of Error**: Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. etc.
- **Discussion**: Write what you have observed and measured in your experiment (as per your understanding)

Experiment 8: Coupled Pendulum

❖ **Objective:** To study the normal modes and resonance of coupled pendula.

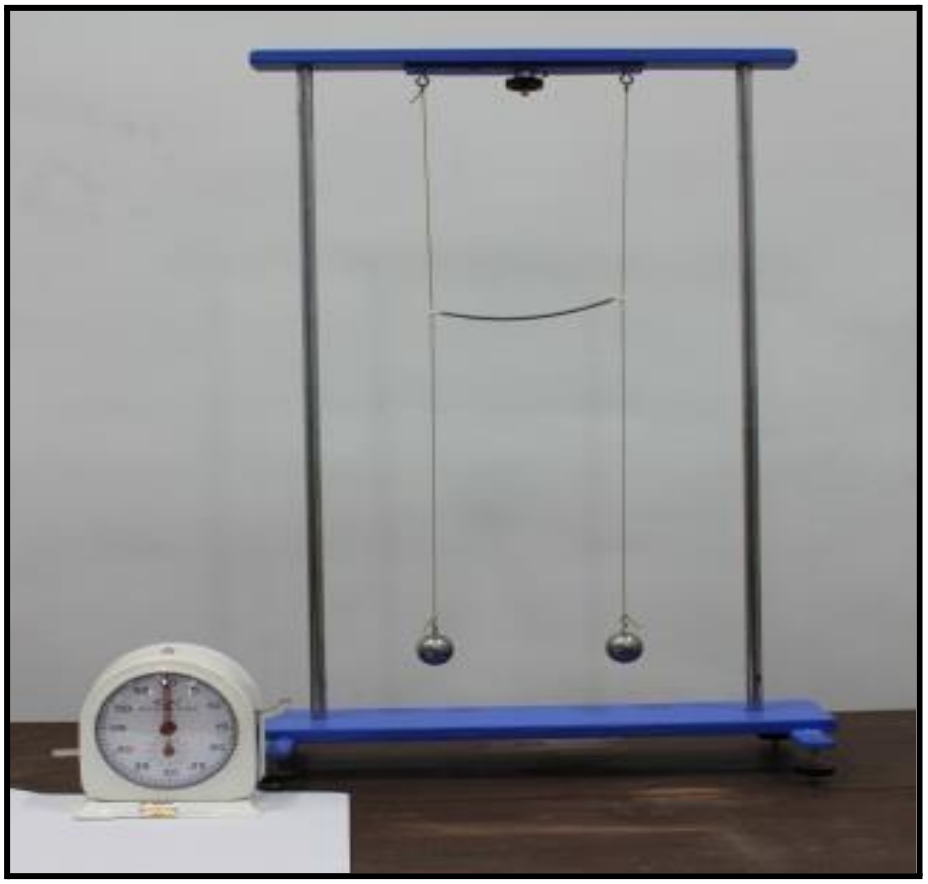
❖ **Apparatus:** Coupled pendulum setup, Spring, Stop watch.

❖ **Task:**

(i) To calculate the time period of the individual pendulum by removing the spring
(ii) Find out the time period of the coupled pendulum when

(a) Vibrating in In-phase mode (T_0)
(b) Vibrating in Out-of-phase mode (T_1)
(c) Vibrating in coupled mode (T_c) and
(d) the Beat period (T_B) and

(iii) Calculate the degree of coupling (D)



❖ **Theory:**

If x and y are the respective displacements of the two masses, then the equations of motions:

$$m\ddot{x} = -\ mg\frac{x}{l} - k(x - y) \quad \text{..... (1)}$$

$$m\ddot{y} = -\ mg\frac{y}{l} - k(y - x) \quad \text{..... (2)}$$

If natural frequency of each pendulum is $\omega_0^2 = g/l$, then we have

$$\ddot{x} + \omega_0^2 x = -\frac{k}{m}(x - y) \quad \ldots\ldots (3)$$

$$\ddot{y} + \omega_0^2 y = -\frac{k}{m}(y - x) \quad \ldots\ldots (4)$$

Adding (3) and (4), we have

$$\ddot{x} + \ddot{y} + \omega_0^2(x+y) = 0 \quad \ldots\ldots (5)$$

Subtracting (3) and (4), we get

$$\ddot{x} - \ddot{y} + \omega_0^2(x - y) + \frac{2k}{m}(x - y) = 0 \quad \ldots\ldots (6)$$

Put (x+y) = X in (5) and (x-y) = Y in (6), we get

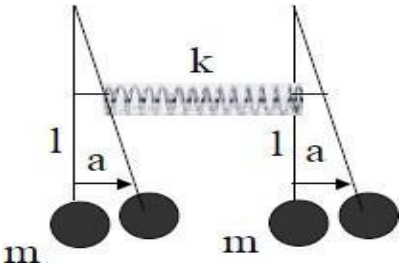
$$\ddot{X} + \omega_0^2 X = 0 \quad \ldots\ldots (7)$$

$$\ddot{Y} + \left(\omega_0^2 + \frac{2k}{m}\right)Y = 0 \quad \ldots\ldots (8)$$

These are equations of S.H.M.

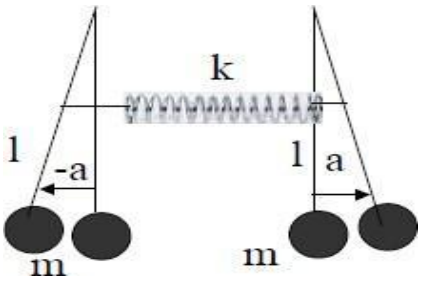
Comparing (7) and (8) with standard equation $\frac{d^2z}{dt^2} + \omega_z^2 = 0$ of S.H.M it is clear that, these are the equation of S.H.M. with natural angular frequencies ω_0 and $\omega_1 = \sqrt{\omega_0^2 + 2\frac{k}{m}}$ respectively.

Case I: **<u>In-phase mode</u>**



If x = y, the motion is completely governed by eq. (7) having frequency of oscillation ω_0.

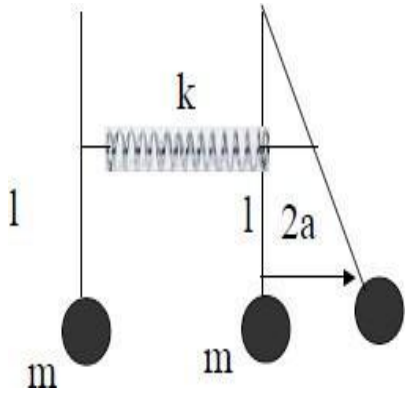
Case II: **<u>Out-of-phase mode</u>**



If x = y, the motion is completely governed by eq. (8) with frequency of oscillation ω_1.

The above two modes are called 'normal modes' of vibration of coupled pendula and the frequencies ω_0 and ω_1 are called 'normal frequencies.

Case III: **Resonance Mode**



In resonance, one mass is at rest and other one move, let say by x=2a.

The general solutions of eq (7) & (8) are the following:

$$X = x+y = X_0 \cos(\omega_0 t + \theta_1) \quad \ldots\ldots (9)$$

and

$$Y = x-y = Y_0 \cos(\omega_1 t + \theta_2) \quad \ldots\ldots (10)$$

For resonance $X_0 = Y_0 = 2a$ and $\theta_1 = \theta_2 = 0$, the displacement of the right pendulum is given by

$$x = \frac{1}{2}(X + Y) = a\cos\omega_0 t + a\cos\omega_1 t = 2a\cos\frac{(\omega_1-\omega_0)t}{2}\cos\frac{(\omega_1+\omega_0)t}{2} \quad \ldots\ldots (11)$$

Similarly, the displacement of the left pendulum is

$$y = \frac{1}{2}(X - Y) = a\cos\omega_0 t - a\cos\omega_1 t = 2a\sin\frac{(\omega_1-\omega_0)t}{2}\sin\frac{(\omega_1+\omega_0)t}{2} \quad \ldots. (12)$$

In resonance two frequencies are involved. One frequency of coupled pendula. We say this frequency as

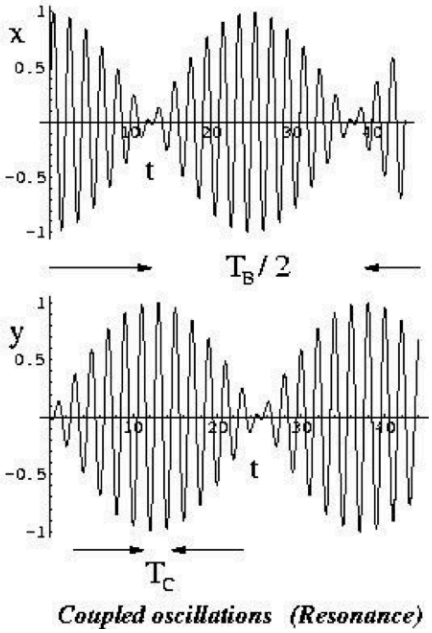
$$\omega_c = (\omega_0 + \omega_1)/2$$

The other frequency component causes the variation in amplitude. This phenomenon is known as beats. Beat frequency is given by

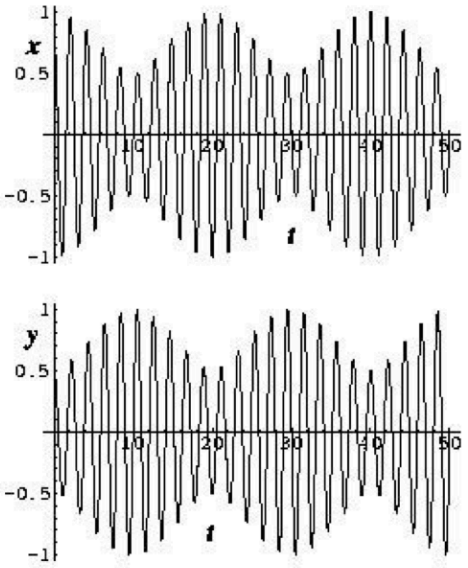
$$\omega_B = (\omega_1 - \omega_0)/2$$

The degree of coupling is defined by

$$D = \frac{\omega_1^2-\omega_0^2}{\omega_1^2+\omega_0^2} = \frac{T_0^2-T_1^2}{T_0^2+T_1^2}$$



Coupled oscillations (Resonance)



Coupled Oscillations

❖ **Procedure:**

1. **Measurement of individual time period (before Coupling or Zero coupling stage)**

(i) Remove the spring and uncouple the two pendula and measure their individual time period. This should be done by measuring the time for 50-100 oscillations. The time period of both pendulums should approximately be the same.

2. **Measurement of In-phase and Out-of-phase mode time periods**

(ii) Now attach the spring between the pendulums to couple them. Make sure that after coupling the pendulums with spring, the distance between the pendulums should be such that there is no stretch in the spring (i.e. the spring should be in its original length) and both the pendulums should be in the same alignment.

(iii) Displace both the pendula equally in the same direction (as shown in Case I) and measure the time period T_0, after measuring the total time t_0 for at least N_0 oscillations.

Now $T_0 = 2\pi/\omega_0 = t_0/$(Total no. of oscillations observed (N_0))……. (In-Phase mode)

(iv) Now, displace both pendulas in opposite directions equally from their respective freely hanging positions (as shown in Case II) and note the time period T_1 after measuring the total time t_1 for at least N_1 oscillations.

i.e. $T_1 = 2\pi/\omega_1 = t_1/$(Total no. of oscillations observed (N_1))…….. (Out-of-Phase mode)

3. **Measurement of Coupled mode and Beat mode time periods (in Resonance Mode)**

(v) Keeping one pendulum at rest, displace the other and let the system settle into performing coupled oscillations. Soon after start measuring the time period T_c of coupled oscillation, after noting down the total time t_c for N_C oscillations (do not touch or hold any of the pendulum).

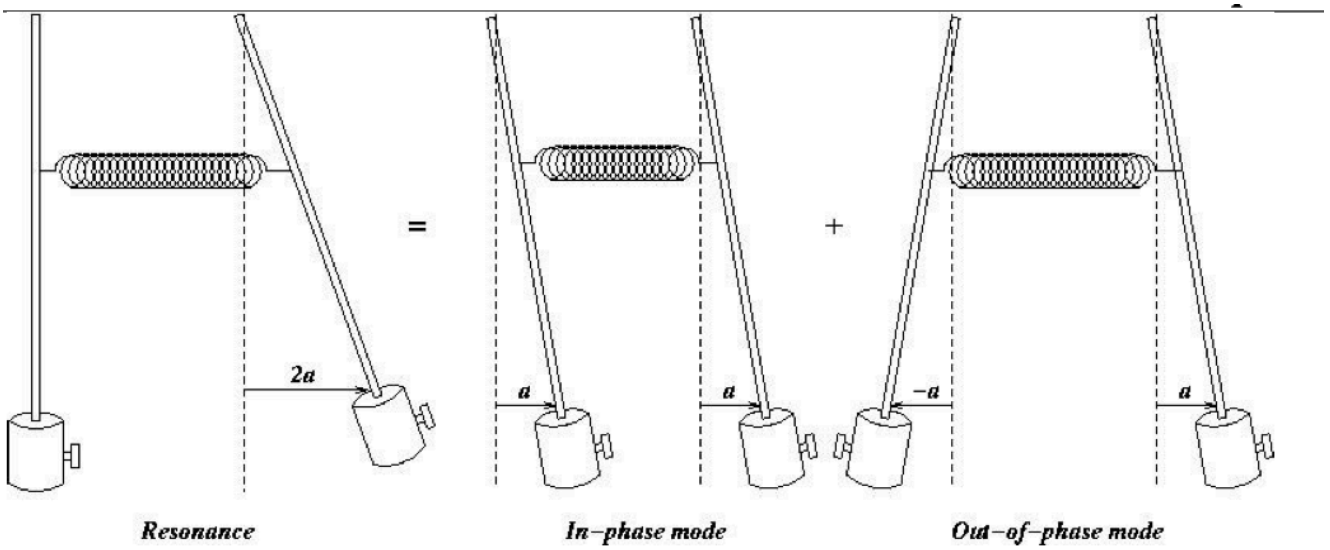
i.e., $T_c = t_c /$(Total no. of oscillations observed (N_C))

(vi) Repeat step (v) as above and now measure the time period T_B (i.e., time interval between two consecutive amplitude minima) for beats by observing only one of the two pendula becoming stationary at least N_B times (do not touch/hold any of the pendulum). The N_B should be more than 3 times. (Refer above figure for beats period and coupled period)

i.e. $T_B = t_B$ /(Total no. of oscillations observed (N_B))

(vii) The values for T_C and T_B are also calculated (using the given formula) and compared with the measured values in steps (v) and (vi).

(viii) The degree of coupling is to be calculated using the given formula.



❖ **Precautions:**

(i)The angular displacement of the pendulum should be kept small (i.e. below 0.3 radian or around ~7 cm from the rest position).

(ii) While using Spring for coupling, check whether it is at its natural length. It should neither be stretched nor loose. If it is too loose, a sag will be produced in the spring, which needs to be avoided.

❖ **Observation:**

Least count of stop watch =

Observation I: Before Coupling (or Zero-Coupling Stage)

Time period of pendulum 1 =

Time period of pendulum 2 =

Observation II: Measuring In-phase and Out-of-Phase modes when coupling is on

Measured Value			Calculated Value		
No. of Oscillations	In Phase mode ($T_0 = t_0/N_0$) (sec)	Out of phase mode ($T_1 = t_1/N_1$) (sec)	$T_c = \frac{2T_0T_1}{T_0+T_1}$	$T_B = \frac{2T_0T_1}{T_0-T_1}$	$D = \frac{T_0^2-T_1^2}{T_0^2+T_1^2}$
35			…….	……..	……..
30			……..	…….	…….
25			……..	…….	……

Observation III: Measuring Coupled mode and Beat mode when coupling is on
(Move Maximum possible error to above table and calculate relative error in below table)

Measured Value				Errors	
No. of Oscillations for Coupled mode	Coupled mode $T_c = t_c/N_c$ (sec)	No. of Oscillations for Beat mode	Beats Mode $T_B = t_B/N_B$ (sec)	Maximum possible error in T_c ($\delta T_c/T_c$)	Maximum possible error in T_B ($\delta T_B/T_B$)
10		4		…….	……..
15		8		……..	…….
20		12		……..	…….

**Take 3 sets of readings for each mode (for* Maximum possible error use the formula given below*)*

❖ **Result :**

(i) Calculate the Time period in coupled mode (T_C)

(ii) Calculate the Time period in Beats (T_B)

(iii) Compare the measured values with the calculated values and find the errors in both T_C and T_B

(iv) Calculate the Degree of coupling (D)

The Degree of coupling is ($D \pm \Delta D$)

❖ **Error Analysis:** Quantify the fractional/percentage error in your results. Do the error calculation using a pen. Show the detailed error calculation in your lab notebook. Discuss with your group members if needed.

Calculate error using differentiation method-

Maximum possible error in T_C

$$\frac{\delta T_C}{T_C} = \left(\frac{1}{T_O} + \frac{1}{T_1} + \frac{2}{T_0 + T_1}\right)\delta T$$

Maximum possible error in T_B

$$\frac{\delta T_B}{T_B} = \left(\frac{1}{T_O} + \frac{1}{T_1} + \frac{2}{T_0 - T_1}\right)\delta T$$

Where $\delta T = \frac{Least\ count\ of\ stopwatch}{No\ of\ oscillations}$

❖ **Source of Error:** Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. Etc.

❖ **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 9: Interference by Newton's Ring

1. **Objective:** To determine the wavelength of monochromatic light (sodium light) using Newton's ring.

2. **Apparatus:** Newton's Ring setup, Plano-convex lens, Plane glass plate, Spherometer.

3. **Task:**

   (i) To determine the diameter of the Newton's Ring.
   (ii) To determine the radius of curvature of the Plano convex lens using a spherometer.
   (iii) Plot the graph between the square of the diameter of rings (D_n^2) and the order of the ring (n), and hence find the wavelength of the light.

4. **Theory/Principle:**

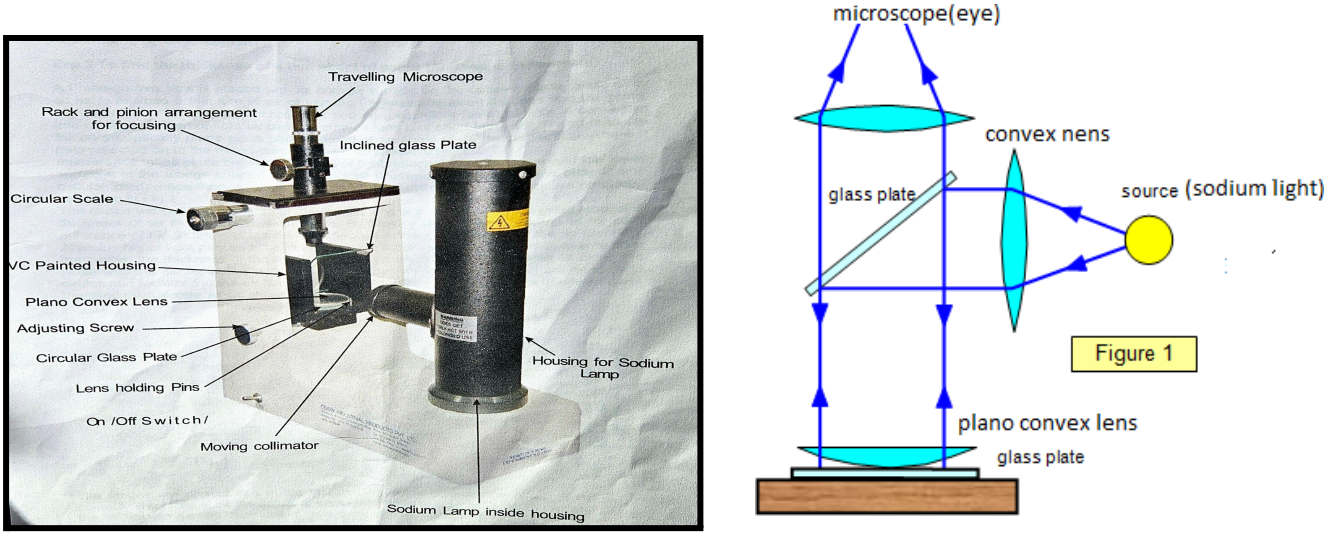


Figure 1:The Experimental Set-up and the schematic ray diagram

As shown in the figure 1, a Plano convex lens is mounted on a glass plate. As the convex surface is in contact with the flat glass plate a wedge-shaped air film of variable thickness is formed. Monochromatic light from an extended light source (sodium lamp) is converted into parallel beam by using a convex lens (figure 1). Then the beam is allowed to fall on an optically plane glass plate inclined at an angle 45 degree to the vertical where it gets reflected on to the Plano convex lens.

Newton's rings are formed due to the interference of light reflected from the upper and lower surface of the wedge shaped air film. Rings can be seen through the travelling Microscope (figure 2).

Let R be the radius of curvature of the convex lens, and D_n be the diameter of the n^{th} Newton's Ring. Then,

$$D_n^2 = 4nR\lambda,$$

where λ is the wavelength of the monochromatic light used.

Figure 2: Circular fringes showing an interference pattern.

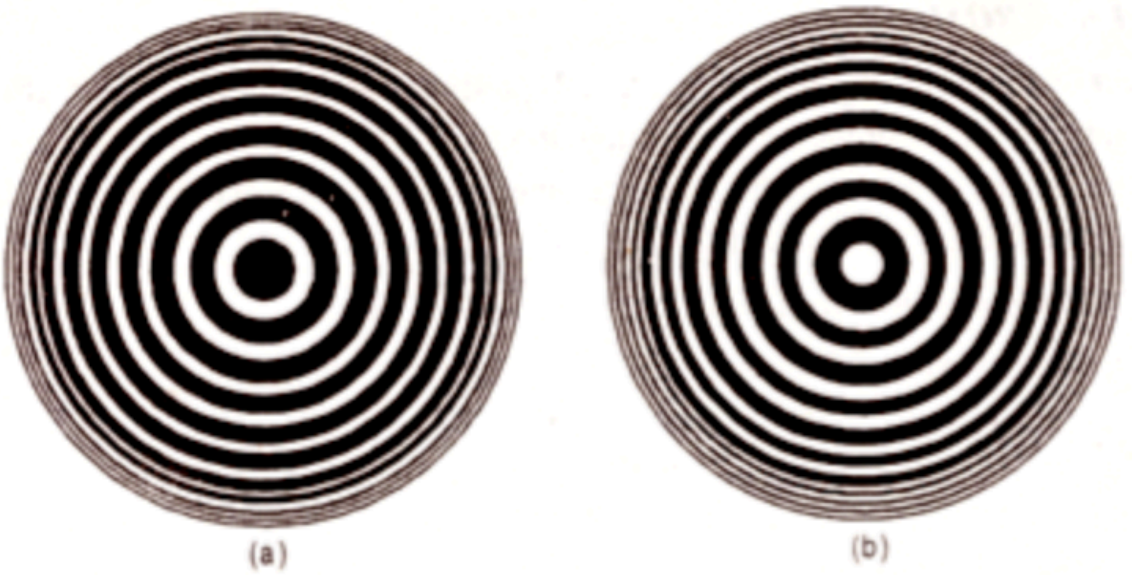


Figure 1 b : fringe pattern for reflected and transmitted light

5. Procedure:

(i) Find the least count of the microscopic scale.

(ii) Clean the surface of the glass plate and the convex lens and place them as stated above (i.e., first put the glass plate then put the Plano convex lens above the glass plate such that the convex surface touches the glass plate).

(iii) Switch on the sodium lamp and wait for 5 to 10 minutes.

(iv) Adjust the microscope to see Newton's rings clearly.

(v) Measure the diameter of different rings. Like, first place the crosswire on the left side of the 20^{th} dark ring. Note the reading from the microscopic scale

(Microscope reading = Main scale reading + Vernier scale reading * Least count)

(vi) Move the cursor towards right to get readings of the left side of 18^{th}, 16^{th}, ……. 4^{th} ring. Then move the crosswire to the other side of the central dark ring to get the reading of the right side of the 4^{th}, 6^{th}…..20^{th}dark ring. Repeat the procedure for the reverse direction (i.e. moving the cursor towards left from the right side of the 20^{th} dark ring).

(vii) Remove the Plano convex lens and find the radius of curvature of its convex surface by using a spherometer.

(viii) Spherometer has three legs (as shown in Figure 3). Measure the distance between the first leg and the second leg (L_1) then between second and the third (L_2) and then first and the third (L_3). Therefore,

$$\textbf{Mean distance (L)} = (L_1+L_2+L_3)/3$$

(vii) The radius of curvature can be determined from the formula

$$R = (L^2/6h) + 0.5h$$

where h is the height at the centre of the convex surface which is calculated from spherometer

readings (as shown in Figure 4).

(viii) Determine the pitch and least count of the spherometer.

Least count = Pitch/Number of divisions on circular scale

(ix) Place the spherometer on a plane glass plate and note the zero reading (b). (Adjust the central screw until all four points (three legs and the screw tip) just touch the glass plate)

(x) Keep the plano-convex lens on the glass plate with the convex surface Upward.

(xi) Place the spherometer on the lens and lower the central screw until it just touches the lens surface, so that all three legs rest symmetrically (as shown in Figure 4).

(xii) Note the reading for plano-convex lens (a) and apply zero correction. The height (h) of the lens is the difference between the reading on the lens and the zero reading.

[h = (Reading on lens) − (Zero reading) = a - b]

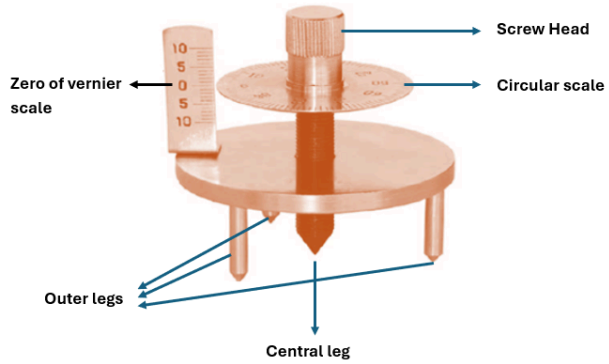


Figure 3: A schematic diagram of a spherometer

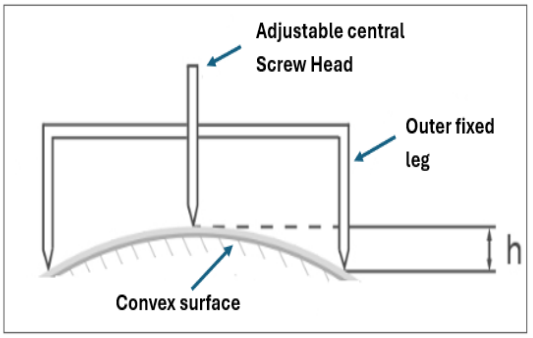


Figure 4: A schematic of determining radius of curvature of a convex surface using spherometer

6. Observation:

Table 1: To determine the diameter of rings (D_n)

Ring no.	Microscope reading crosswire going from left to right		Diameter (b-a) or (a-b)	Microscope reading crosswire going from right to left		Diameter (c-d) or (d-c)	Mean diameter (D_n)
	Left (a)	Right (b)		Right (c)	Left (d)		
16							
14							
12							
……	…..						

**Take readings for 8 rings.*

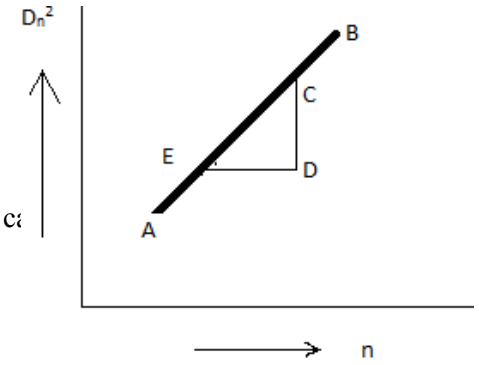
Table 2: To determine the height (h) of the centre of the convex lens

Sr.no.	Spherometer reading		h = a-b; if a > b h = 100 + (a-b); if a < b
	Convex surface (a)	Plane surface (b)	
1			
2			
3			
		Mean h = ..	

Table 3: To plot D_n^2 vs n graph

Order of ring (n)	D_n	D_n^2
1		
…		

7. Calculation of the wavelength using the graph:



Slope of a straight line = CD/DE

From the relation $D_n^2 = 4nR\lambda$, we get slope = $4R\lambda$

Thus CD/DE = $4R\lambda$, hence one can easily

8. Result:

(i) Plot the graph and find out the wavelength of the sodium light (λ)
(ii) Find out the error in your measurement.
(iii) The wavelength of the sodium light is ($\lambda \pm \Delta\lambda$) meter

❖ **Error Analysis**: Quantify the fractional/percentage error in your results. Do the error calculation using a pen. Discuss among your group members if needed. Theoretical value of λ = 589 nm (Given).

❖ **Source of Error:** Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. Etc.

❖ **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 10: Polarimeter

- **Objective**: To determine the specific optical rotation of a sugar solution.
- **Apparatus:** Polarimeter, Test tube, Sugar, Distilled water, weighing machine

Figure 1: Experimental set-up

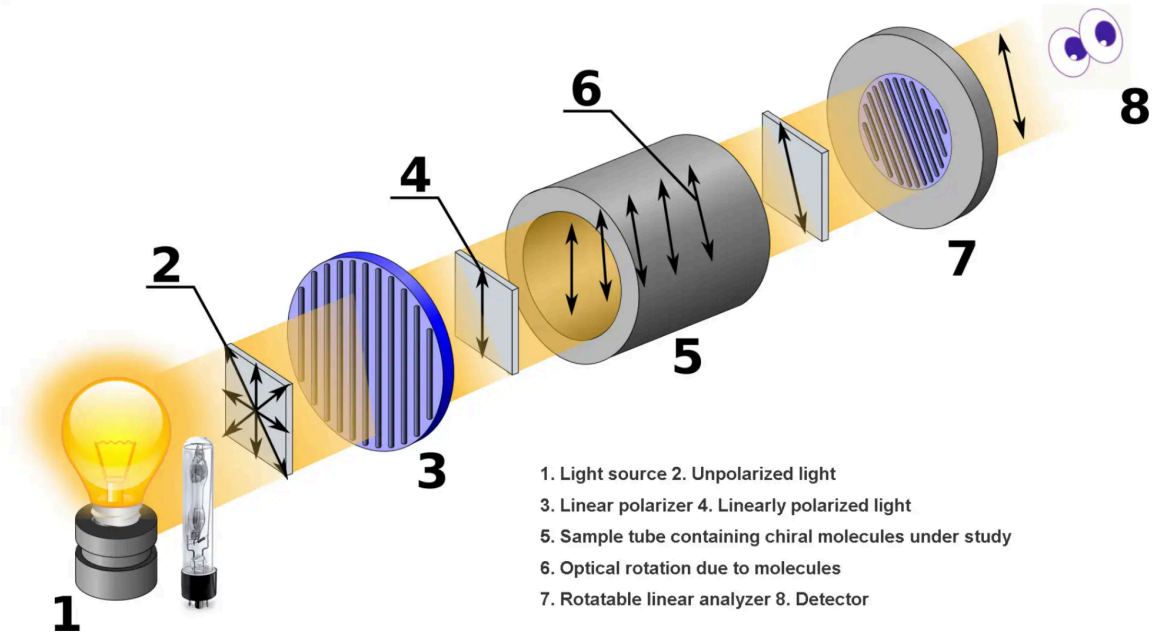


Figure 1b: principle of operation of polarimeter

❖ **Task:**

(i) To calculate the specific rotation (s or α) for different concentrations of sugar solution
(ii) To draw the calibration curve (concentration vs angle of rotation (θ))

❖ **Principle**: The principle of operation of the Polarimeter is established on the basis of the polarized light (shown in Figure 1b). In addition, by use of the method of rotating the polarized plane of the polarization light, its measuring purpose can be attained. At the position of zero degree, AA' is perpendicular to the centre line OX (as shown in figure 2)

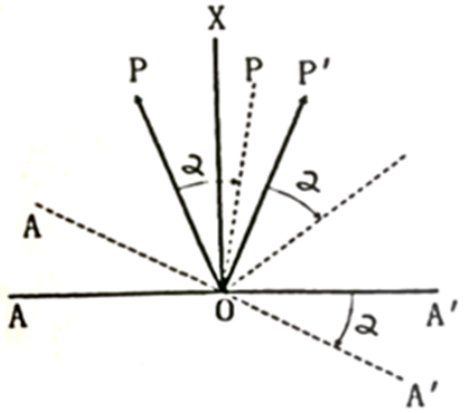


Figure 2: Schematic ray diagram for studying the specific optical rotation of a sugar solution

AA' indicated the vibration direction of the polarization analyser and OP, OP' indicate the vibration directions of the polarized lights of the two halves of the viewing field. When the light beam passes through the optical rotators substance, the plane of polarization is rotated by an angle α, as shown by the dotted line in the above figure. Now, the projection of the polarized lights of the two halves on AA' are not identical, the right half is bright and the left half is dark.

If rotating the polarization plane AA' of the analyser by an angle of α in the same direction, the illumination intensity of the viewing field can be made equal again: At this time the rotated angle of polarization analyser is the optical rotation of the substance. Knowing the rotated angle (i.e. the optical rotation), the length of sodium column (i.e. the length of the test tube) and the concentration, the specific optical rotation of the substance can be calculated in accordance with the following formula:

$$S = \frac{\theta}{l\,C}\ x\ 100$$

where θ is rotation angle (optical rotation) measured by the use of light λ, when the temperature is T. *l* is the length of the solution column (test tube) using decimetre as the unit of length.
C is the concentration, i.e., the gram quantity of the solute in 50 ml of solution

Observing through the eye lens we can see a mid-bright (or dark) and left/right dark (or bright) triplex viewing field of unequal intensity of illumination (see figure 3(a) and figure 3(b)).
Rotate the dial rotary hand wheel so as to drive the dial and the polarization analyser until the illumination intensity of the viewing field (dark viewing field) becomes equal (see figure 3 (c). Then the angle of dial rotation can be read out from the magnifying glass (see figure 4)

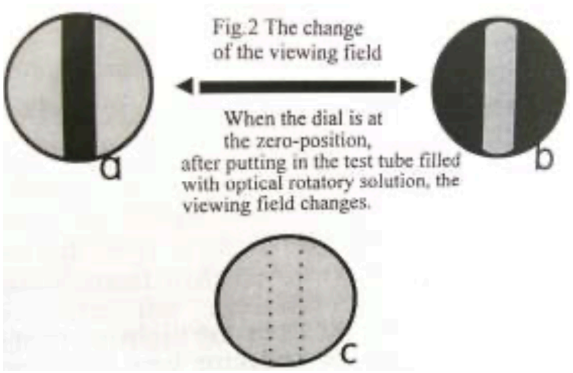


Figure 3

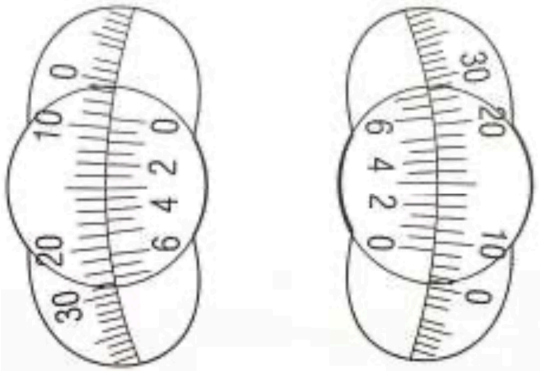


Figure 4

❖ **<u>Procedure</u>:**

(i) Compound the solutions to be measured and then have the solution stabilized.

(ii) Pour the solution to be measured into the test tube for measurement. Caution: The screws at the both ends of the test tube should not be screwed in too tightly (usually screw down the screws with hand conveniently until no water leak occurs)

(iii) Connect the power source and turn on the sodium lamp, and wait for 5 to 10 minutes. Only when the lamp gives out yellow sodium light can the observation be made.

(iv) Open the cover of the lens cone, and put the test tube into the lens cone for measurement. Then, close the cover of the lens cone and have the bulb part of the test tube upward, so that the air bubbles can be stored in the bulb part and the observation and measurement will not be affected.

(v) Adjust the viewing field screw till the triplex viewing aspect in the viewing field becomes distinct.

(vi) Rotate the dial hang-wheel until the illumination of the viewing field (the dark viewing field) becomes identical.

(vii) Read out the rotated angle by the dial through the magnifying lens attached.

(viii) Now replace the test tube with different concentrations and repeat steps 5, 6, and 7.

(ix) Calculate the specific optical rotation for different concentration solutions.

❖ **<u>Observation</u>:**

Length of the tube =

Least count of the Polarimeter =

Table for calibration curve

S. No.	Concentration (C)	Angle of rotation (θ)
1		
2		
……	……	

**Take reading for 5-7 different concentrations*

❖ **Results:**

(i) Plot of C vs θ [use the best fit]

(ii) Find out the value of specific rotation (S) from the graph using the best-fit curve/line

[Note: You have to take a point P on that graph with arbitrary C and its corresponding θ to calculate S]

The specific rotation should be reported as $(S \pm \Delta S)$

❖ **Error Analysis:** Quantify the fractional/percentage error in your results. Do the error calculation using a pen. Discuss among your group members if needed.

Theoretical value of S for sugar = 66^0

❖ **Source of Error:** Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. etc.

❖ **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 11: Four probe

9. **Objective**: To measure the energy band gap of a given semiconductor using Four probe method.

10. **Apparatus:** Four probe set-up, Oven, Connecting leads, Thermometer sensor.

11. **Task:**

    (i) To record probe voltage for different values of temperature for a semiconductor.
    (ii) To plot the resistivity versus the inverse of temperature graph.
    (iii) To calculate the energy band gap of the semiconductor provided.

12. **Principle:** The purpose of the 4-point probe method is to measure the resistivity of a semiconductor sample. The 4-probe setup consists of four equally spaced metal tips with a finite radius. Each tip is supported by springs on the other end to minimize excessive pressure on the crystal. A high impedance current source is used to supply current through the outer probes; a voltmeter measures the voltage across the inner two probes to determine the sample resistivity and hence, the energy band gap of the semiconductor sample.

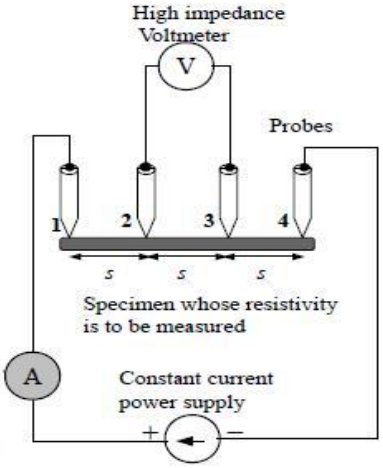


Figure 1a: Schematic diagram of four-probe measurement.

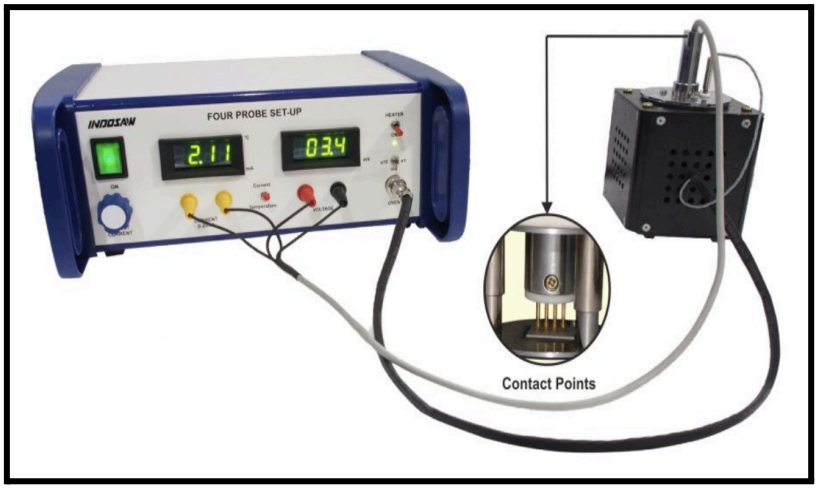


Figure 2: Picture of four probe setup

Four sharp probes are placed on a wafer (semiconductor sample), current is passed through the two outer probe and the floating potential is measured across the two inner probes to determine the sample resistivity.

The energy band gap, E_g of a semiconductor, is given by

$$E_g = (2k_B\ 2.3026 \times \log_{10}\rho) / T^{-1}\ (\text{in eV})$$

where, $k_B = 8.6 \times 10^{-5}$ eV/K

ρ is resistivity of semiconductor sample given by $\rho = \rho_0 / f(W/s)$

where, $\rho_0 = (V/I)*2\pi s$,

W is thickness of sample, s is the probe spacing, f(W/s) function is the correction factor, V is the voltage across the two inner probes, and I is the current through the two outer probes.

13. **Procedure:** (i) The p-type Ge sample is factory fitted. If there is any need for replacing or mounting a sample then, proceed as per following procedure. Unscrew the pipe of the four probe arrangement and place the sample on the base plate of 4-probe arrangement and adjust the pipe such that the 4-point probe lie in the middle of the sample as shown in figure 3.

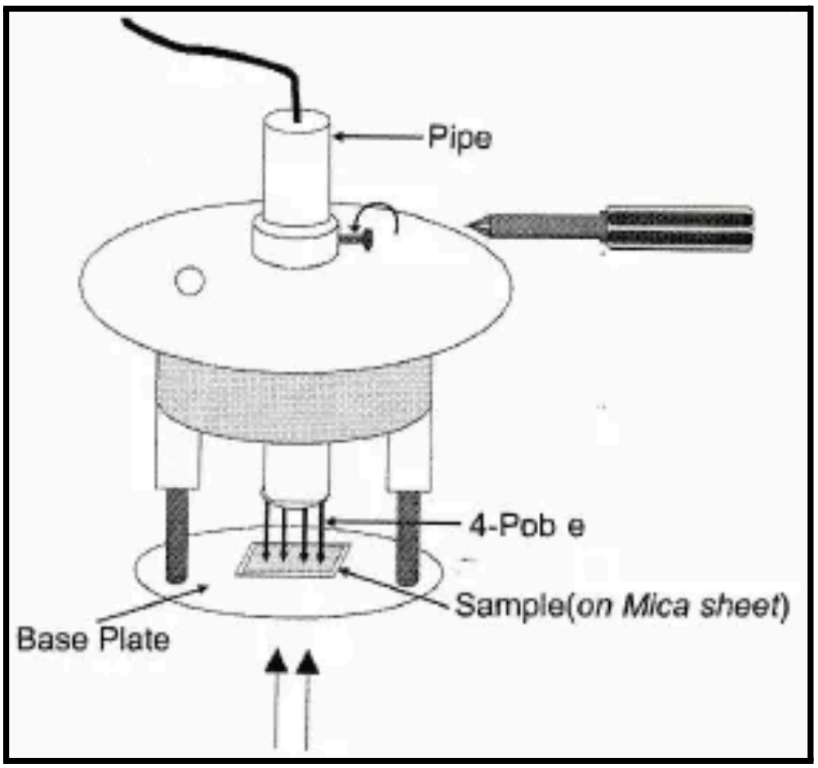


Figure 3: Shows four probe arrangement for mounting the sample.

Gently apply some pressure and tighten the pipe in this position such that all the probes are in contact with the sample. {Warning! Applying excessive pressure may break the sample.)

(ii) Put the four-probe arrangement in the oven and connect the lead of the oven to the socket. Also, insert the PT-100 temp sensor into the hole given at the top of the 4-probe arrangement.
(Note: Do not switch 'ON 'the oven. Keep the switch of the oven in the OFF position.)
(iii) Connect the RED and BLACK plug leads of 4- probe arrangement to the 4 mm sockets marked as "Voltage".
(iv) Connect the YELLOW plug leads to 4mm sockets marked as "Current".

(v) Change switch to current, so display show current reading.
(vi) Switch 'ON 'the apparatus. Slightly increase the current using current knob, say 4mA and note that the voltage should be positive. If it is not, interchange the current leads.
(vii) Set the current to desired value (say 8 mA) using current adjusting knob. Also select the range of multiplier using switch to X1 or X10 on voltage display. (It is always better to start with lower range i.e. Xl)
(viii) Switch 'ON 'the oven using switch. Green LED will glow, showing that the oven is ON.
(ix) Change switch to temperature, so display show temperature.
(x) Note the probe voltage on display for different values of temperature as shown on display.
(xii) Take readings between (30-150 ^{o}C) temperatures in steps of 5 ^{o}C.

14. Observation:

a. Current, I = 8.03 mA (constant)
b. Distance between probes, s = 0.24 cm
c. Thickness of sample, W = 0.05 cm

S. No.	Temperature(^{o}C)	Voltage(mV)	Temperature(K)
1	30		
2	35		
3	……		
…….	150		

**Take 25 set of readings (more data points in high temperature region)*

Find the resistivity corresponding to different temperatures using the expression below -

$\rho = \rho_0 / f(W/s)$ & $\rho_0 = (V/I)*(2\pi s) =$ …….. Ω cm

[*Note: For different values of V, there will be different values of ρ_0*]

where, f (W/s) can be found from the below table.

W/s	f(W/s)
0.100	13.863
0.141	9.704
0.200	6.931
0.333	4.159
0.500	2.780
1.000	1.504
1.414	1.223
2.000	1.094
3.333	1.0228
5.000	1.0070
10.000	1.00045

[Note: if any W/s and its corresponding f (W/s) are not found in the table, then the value can be found by using interpolation.]

Now, calculate the value of resistivity ρ for different values of ρ_0 and make an observation table

S.No.	$T^{-1} \times 10^3$ (K^{-1})	ρ (Ω cm)	$\log_{10} \rho$
1			
3			
…….	……..	…….	…….

Plot a graph (T^{-1} x 10^3) versus $\log_{10}\rho$ using the observation table and find the slope of the curve as shown in a typical graph (Figure 3)

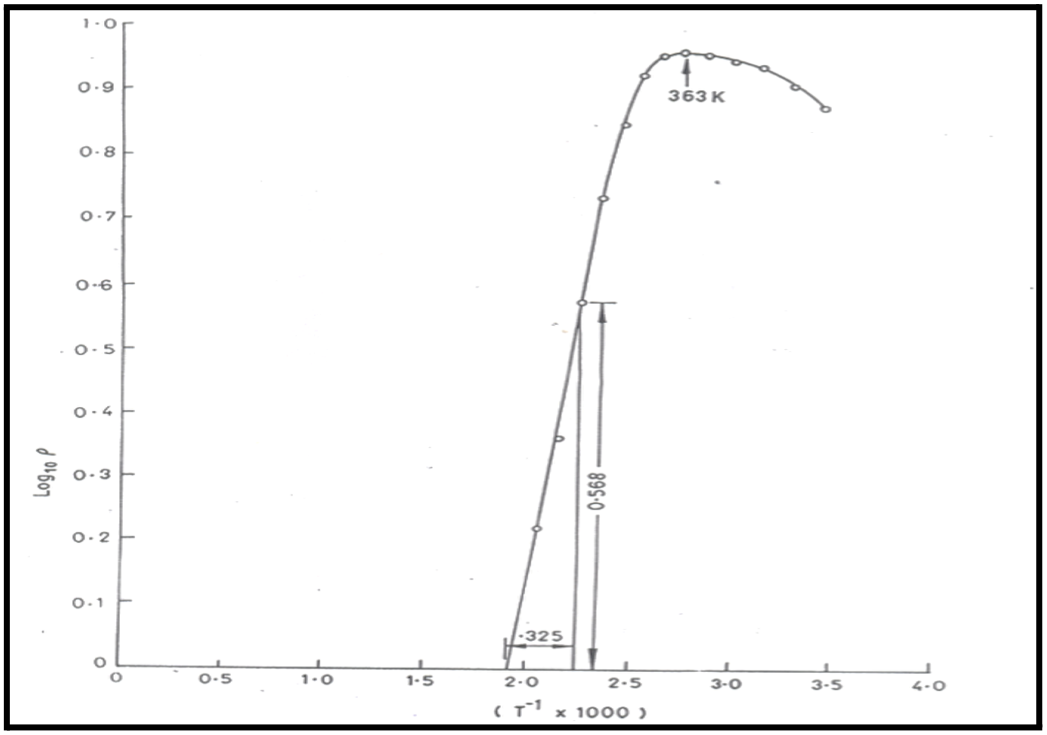


Figure 3: Graph of resistivity of a semiconductor crystal as a function of inverse temperature

From the above graph, find the slope in the linear region of the curve by drawing the slope (a linear line) and use the slope value in the given formula of energy band gap.

Result:

(i) Plot Resistivity vs Temperature graph.
(ii) Find out the value of energy band gap.

The energy band gap is ($E_g \pm \Delta E_g$) eV

Error Analysis: Quantify the fractional/percentage error in your results. Do the error calculation using a pen. Discuss among your group members if needed.

Theoretical value of E_g = 0.7 eV (Germanium)
= 1.12 eV (Silicon)

- **Source of Error**: Discuss the source of error with your group members and write about it. These errors should be specific to your instrument and the experiment. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. Etc.

- **Discussion**: Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.

Experiment 12: Diode laser diffraction

15. **Objective:** To determine the wavelength of diode laser light using diffraction phenomena

16. **Apparatus:** Optical bench, Diode laser, Digital multimeter, Diffraction grating, Pin hole photo detector

17. **Task:**

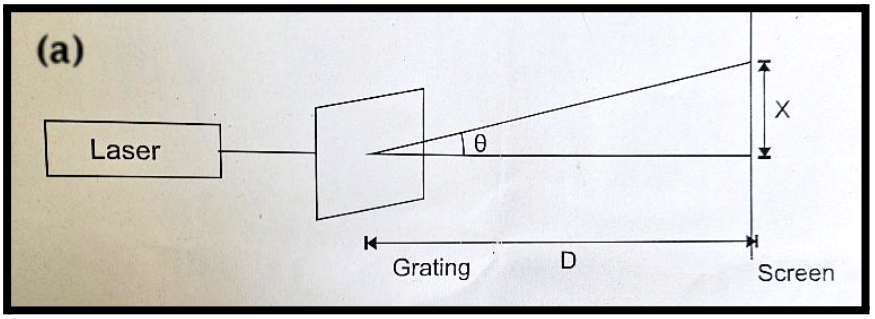
(i) To plot the intensity versus distance curve
(ii) To calculate the wavelength of the diode laser using the diffraction grating
(iii) To calculate the width of the single slit

18. **Principle:** The phenomenon which make it evident that light under certain circumstances (i.e. when size of obstacle is comparable with wavelength of waves) does not travel in straight line is called as diffraction. Diffraction of light occurs as it passes through a slit, the angle to the minima in the diffraction pattern is given by

$$a \sin\theta = m \lambda \quad \ldots\ldots\ldots\

(ii) A plane transmitting grating is mounted on an upright next to the laser.

(iii) On the other end put a screen with white paper on it.



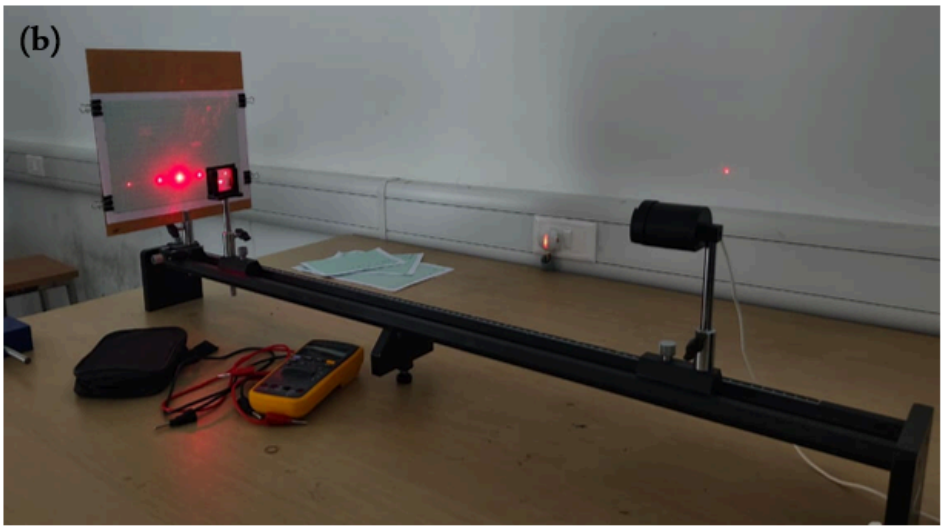


Figure 2: Determine the wavelength of the light source:(a) Schematic diagram, (b) Experimental set-up

(iv) The position of the spot of 1st order on either side of the central maxima is marked x.

(v) Calculate the $Sin\ \theta = \frac{x}{\sqrt{x^2+D^2}}$

(vi) Grating element is d.
It can be calculated as for example for 80 lines per mm, **d** = $\frac{1}{80}\ mm = 12.5\ \mu m$

(vii) Then calculate the wavelength for different distances from the screen.

Formula to calculate the wavelength,

$$\boldsymbol{\lambda} = \frac{d\ Sin\theta}{m} \quad \text{.....(2)}$$

Where **d** is grating constant and **m = 1, 2…** (Order of spectra)

The wavelength of the diode laser is ($\lambda \pm \Delta\lambda$) meter

20. Observation:

1. Measuring the wavelength of the light source

Table 1:

Screen Distance	Order of spectra	Distance (x)	$Sin\,\theta = \frac{x}{\sqrt{x^2+D^2}}$	λ (nm)
D_1 = … cm	m=1			
	m=2			
D_2 = … cm	m=1			
	m=2			
D_3 = … cm	m=1			
	m=2			

2. Measurement of slit-width of the given single slit

In the single slit experiment, the angles are usually small, hence the formula a $\text{Sin}\theta = m\lambda$ can be approximated for small angles. You assumed that $\text{Sin}\theta \sim \tan\theta$ and $\tan\theta \sim y/D$, where y is the distance on the screen from the centre of the pattern to the mth minimum and D is the distance from the slit to the screen as shown in figure 3.

Therefore the slit width :

$$d = \frac{m\lambda D}{y} \quad (m = 1, 2, 3.....) \quad \ldots\ldots\ldots\ldots\ldots\ldots (3)$$

(i) Set up the diode laser at one end of the optics bench and place the slit holder about 5-10 cm in front of the laser

(ii) Put the photodetector at the other end of the optical bench. Adjust the vertical and horizontal position and the laser and detector so that the beam falls at the center of the pinhole.

(iii) Connect the multimeter to the detector. You may take the reading for ambient light before switching on the laser light. This can be noted as zero offset, which can be subtracted later to get the final intensity value.

(iv) Mount the single slit in the slit holder and adjust the position of the laser beam from left to right and up and down until the beam is centered on the slit (and intensity is maximum).

(vi) Determine the distance from the slit to the detector (D).

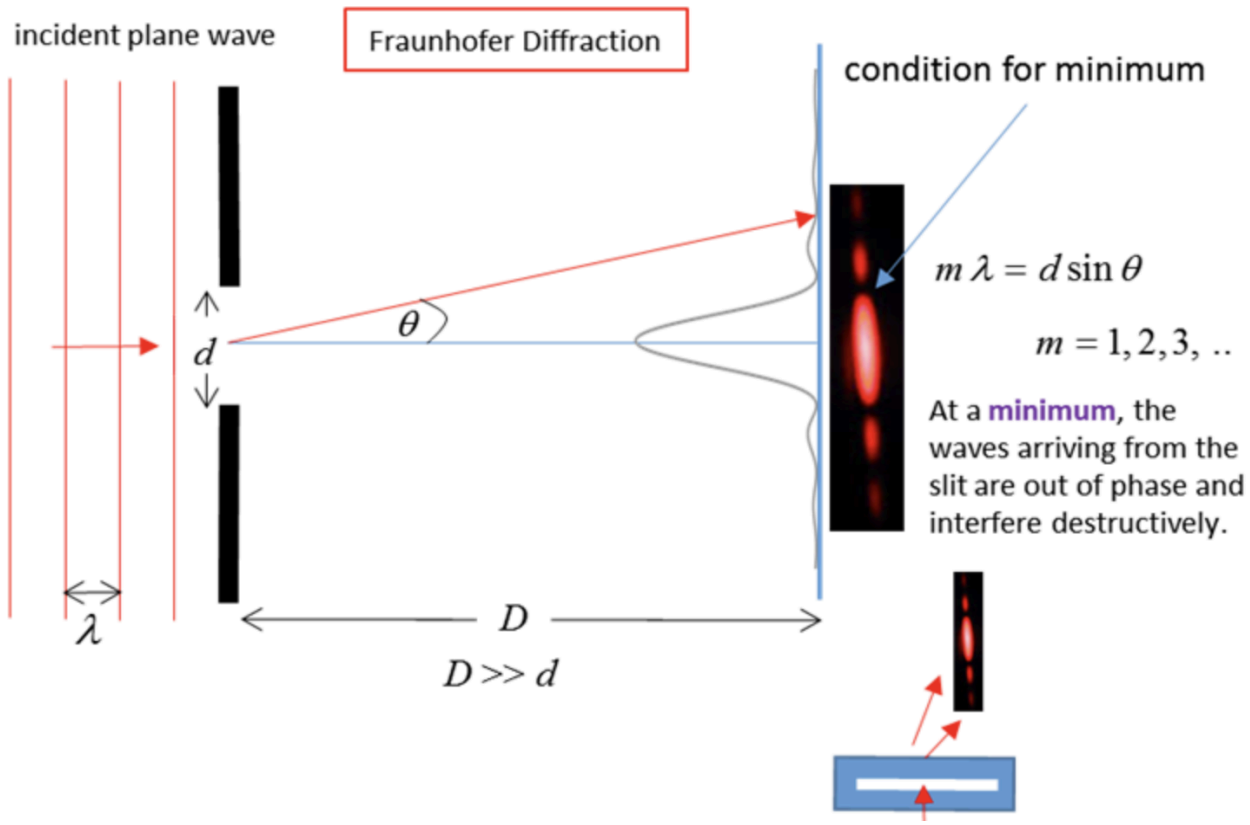


Figure 3: Schematic diagram of the intensity distribution across the screen

(v) Using a micrometer screw (transversal saddle), move the horizontal detector position (y) to the right end (and left) of the diffraction pattern and record the intensity at closed intervals by traversing the detector through the cross-section of the spectrum. (Before taking readings, make sure that you can traverse through both sides of the central peak and record enough readings for 2-3 minima). You may also use a vernier caliper to measure the distance (y) with respect to the reference point.

(vi) Plot the intensity versus distance graph (as shown in Figure 3).

(vii) Calculate the slit width 'd' for mth order (wavelength of the diode laser is given). Calculate for each mth order and then average it.

21. **Observation:**

Distance between the slit and the photo detector, D =

Table 2: Intensity versus distance

S. No.	Distance (y) in mm	Intensity(I) in mV
1		
2		
…….	……..	…….

**Take 10-20 set of readings*

Table 3: Find the width of the single slit using the data collected in Table 2 and using Equation 3

S. No.	Order of spectra m (right)	Distance (y) (mm) (right)	Slit width $d = \frac{m\lambda D}{y}$ (mm) (right)	Order of spectra m (left)	Distance (y) (mm) (left)	Slit width $d = \frac{m\lambda D}{y}$ (mm) (left)	Slit width d (mm) (average)
1	1			1			….
2	2			2			….
						Mean Slit width =	

Graph:

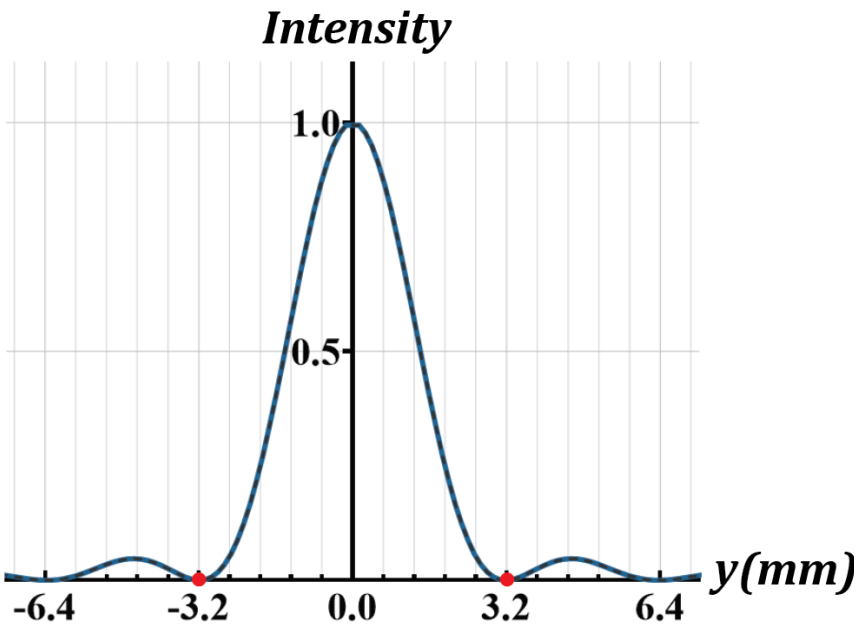


Figure 3: A typical graph (for illustration, not to scale) showing the intensity vs distance

22. **Result:**

(i) Calculate the wavelength of the diode laser using the diffraction grating. Compare results with the given theoretical value.

(ii) Calculate the width of the single slit.

(ii) Plot Intensity vs Distance Graph.

The width of the single slit is ($d \pm \Delta d$) mm

- **Error Analysis**: Quantify the fractional/percentage error in your results. Do the error calculation using a pen. Discuss among your group members if needed.

Theoretical Value of Red diode Laser wavelength = 635 nm

- **Source of Error:** Discuss the source of error with your group members and then write about it. These errors should be specific to your instrument and the experiment. Write down the least count of each instrument you used to do the measurement. Avoid writing generalized terms such as 'Instrumental error' or 'Human error',.. etc.
- **Discussion:** Write what you have observed (as per your understanding). Qualitatively address the amount of error incurred in the experiment, discuss the various results you obtained from this work, and importantly what you learned from this experiment.